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Enhancing Consent Policy Privacy in Blockchain-Based Healthcare Data Sharing

Kun Li [1,*], Ankur Lohachab [1], Michel Dumontier [1] and Visara Urovi [1]

[1] Department of Advanced Computing Sciences, Maastricht University, 6200 MD Maastricht, The Netherlands; ankur.lohachab@maastrichtuniversity.nl (A.L.); michel.dumontier@maastrichtuniversity.nl (M.D.); v.urovi@maastrichtuniversity.nl (V.U.)

* Correspondence: k.li@maastrichtuniversity.nl

Abstract

Public-chain consent enforcement in healthcare data sharing can expose governing policies through transaction artifacts even when health data are encrypted. Hiding policy logic alone leaves policy-dependent metadata observable. To address this gap, we present the Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter (TPS-ZKPI), which delegates private proof generation to a trusted service while verifying compliance with committed, bounded Boolean consent policies on-chain. Fixed-length encoding, commitment binding, and a reusable interpreter make the proof interface policy-independent, while transcript normalization limits variation in calldata, event shape, and gas for the evaluated policies. We assess the resulting public transcript on Polygon Amoy using generated policies and supervised and unsupervised attacks that model a passive public-chain metadata observer. Of 45 supervised tasks, 43 stay below our multi-seed leakage criterion. Diagnostics link the two threshold crossings to chance correlations between policy labels and independently drawn padding targets. A paired ablation further shows that padding suppresses policy-correlated gas signals on the evaluated subset, with mean gas overheads of about 46% for access and 38% for policy commitment. Together, these results support reduced leakage through the modeled metadata channels within the tested policy class and under the trusted-service assumption.

Keywords: blockchain; consent policy privacy; zero-knowledge proofs; Data Use Ontology (DUO); healthcare data sharing; metadata leakage

1. Introduction

Blockchain technology has emerged as a promising foundation for decentralized and auditable healthcare data sharing [1–3]. Its transparency and immutability enable verifiable consent enforcement and traceable authorization decisions, supporting cross-institutional accountability. At the same time, consent enforcement on public blockchains may expose policy-dependent contextual information, even when the underlying health data remain encrypted [4,5].

In healthcare data sharing, a *consent policy* encodes the conditions under which a data subject authorizes others to access or use their health data, such as who may access the data, for what purpose, and under what constraints. Consent policies in healthcare are often privacy-sensitive [6]. For example, a policy based on biomedical data-use terms may reveal whether a dataset is restricted to disease-specific research, whether ethics approval is required, or whether access is limited to particular institutions or projects. If such policies are directly represented in smart contracts, embedded in verification logic, or reflected

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in public execution traces, unauthorized observers may infer information about the protected resource, the data subject's constraints, or the requester's authorization context.

Existing privacy-preserving blockchain solutions mainly protect data contents, user identities, or transaction details [7–9]. Encrypted data-sharing mechanisms such as attribute-based encryption (ABE) and proxy re-encryption protect access to data [10,11], while zero-knowledge transaction systems hide asset ownership, values, or participant relationships [12]. None of these mechanisms protects the confidentiality of the policies that govern access decisions. Even mechanisms designed to hide policy content, such as hidden-policy attribute-based encryption [13,14], do not control the policy-dependent artifacts that a public chain exposes while enforcing the policy. In blockchain-based consent management, policy leakage may arise not only from explicit policy disclosure, but also from policy-dependent artifacts such as calldata size, public-input layout, emitted events, verifier interfaces, and gas usage. Therefore, policy-private consent enforcement requires more than hiding data or proving correctness in zero knowledge; it also requires controlling what the public-chain transcript reveals about the hidden policy. This paper studies this problem for blockchain-based healthcare data sharing under public-chain observability. The central challenge is to enable an on-chain verifier to check that a requester satisfies a provider-defined consent policy while ensuring that metadata-only public-chain observers learn minimal information about the governing policy and the requester's authorization evidence from the resulting verification transcript. We focus on Boolean-based consent policies, including policies derived from biomedical data-use conditions such as terms from the Data Use Ontology (DUO), the Global Alliance for Genomics and Health (GA4GH) standard vocabulary of consent-based data-use conditions [6], and disease-scope constraints [15].

We address this problem with the Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter (TPS-ZKPI). In TPS-ZKPI, the data provider commits to a hidden consent policy on-chain and provisions the corresponding policy-bound proving material to a Trusted Proof Service (TPS). The requester submits policy-relevant authorization evidence, such as attributes, credentials, and access context, to the TPS. The TPS constructs the private witness and generates a zero-knowledge proof showing that the requester satisfies the committed policy. The proof is then verified on-chain against the provider's policy commitment. This yields an auditable authorization outcome without exposing the policy logic to the public-chain transcript.

TPS-ZKPI combines the following design elements. First, it uses TPS-assisted proof generation to keep confidentiality-sensitive policy material and requester evidence outside the public-chain transcript. Second, it compiles heterogeneous Boolean-based consent policies into fixed-length Reverse Polish Notation (RPN) programs. Third, building on the standard commit-and-prove paradigm, it evaluates these programs through a reusable zero-knowledge policy interpreter that avoids policy-specific verifier contracts and supports policy evolution without changing the public verification interface. Fourth, it applies observability-aware transcript normalization to reduce policy-dependent variation in calldata, logs, public parameters, and gas-related behavior.

We evaluate TPS-ZKPI using attack-driven empirical analysis under a metadata-only threat model. The evaluation operationalizes policy privacy through three objectives: Policy Content Hiding (PCH), Policy Indistinguishability (PI), and Structure and Parameter Privacy (SPP). Supervised and unsupervised attacks test whether observable public-chain artifacts allow adversaries to recover hidden policy content, distinguish otherwise-equivalent policies, or infer structural and parameter properties. After transcript normalization, 43 of the 45 supervised tasks stay below a multi-seed leakage criterion; the two that cross it are carried by the random padding targets. A paired ablation serves as a posi-

tive control: without gas padding, the same attacks recover a policy-correlated gas signal, and randomized padding suppresses it.

This paper makes the following contributions:

- We propose TPS-ZKPI, a TPS-assisted zero-knowledge approach for publicly verifiable and auditable consent enforcement, in which on-chain proof verification and auditability hold without trusting the TPS, while policy confidentiality and faithful witness construction are conditional on an explicit, delimited trust assumption—keeping policy-bound proving information and requester authorization evidence off-chain.
- We develop a reusable zero-knowledge policy interpreter for fixed-length Boolean RPN policy programs over a bounded token alphabet, combining policy commitment binding with a policy-independent on-chain verification interface that serves all policies through one circuit and supports policy evolution without changing the on-chain verifier.
- We provide an attack-driven empirical evaluation of residual policy-dependent leakage from normalized public-chain metadata on the Polygon Amoy testnet[1], using supervised classification and unsupervised clustering attacks under a multi-seed permutation-test criterion. A paired gas-padding ablation then attributes the suppression of the modeled gas channel to randomized gas padding and quantifies its gas-cost overhead.

The remainder of this paper is organized as follows. Section 2 reviews the technical foundations: cryptographic access control in blockchain healthcare, zero-knowledge proofs, and the Data Use Ontology. Section 3 discusses related work and positions TPS-ZKPI against existing blockchain consent-management, privacy-preserving blockchain, hidden-policy, private policy-evaluation, reusable proving, and metadata-leakage approaches. Section 4 defines the system and trust model, the public-chain adversary, and the privacy objectives. Section 5 presents the TPS-ZKPI design. Section 6 reports the empirical evaluation. Section 7 discusses limitations and future work, and Section 8 concludes the paper.

2. Background

This section reviews the technical foundations relevant to policy-private consent enforcement on blockchains. We first introduce the cryptographic access-control mechanisms commonly used in blockchain-based healthcare data sharing. We then review zero-knowledge proofs and the Data Use Ontology (DUO) [16], which together provide the basis for the policy representation and verification mechanisms used in our approach.

2.1. Cryptographic Access Control in Blockchain Healthcare

Fine-grained consent management in blockchain-based healthcare data sharing is commonly built on cryptographic access-control primitives, chiefly Ciphertext-Policy Attribute-Based Encryption (CP-ABE) and Proxy Re-Encryption (PRE) [11,17,18]. In CP-ABE, the encrypting party embeds an attribute-based access policy into the ciphertext, so that only requesters whose credentials satisfy the policy can decrypt the protected data [10]. PRE lets a semi-trusted intermediary transform a ciphertext encrypted for one party into one decryptable by a designated requester, delegating access without exposing the plaintext [18]. Both primitives remain the building blocks of current blockchain healthcare designs: recent schemes combine PRE with on-chain records to delegate access to medical records and Internet of Medical Things (IoMT) data [19,20], and CP-ABE variants hide the access policy and attributes [13,14], in one case also replacing the sin-

[1] https://amoy.polygonscan.com

gle trusted authority with distributed key generation [14]. These primitives establish who may obtain the data; how they relate to *policy* confidentiality on a public chain is discussed in Section 3.

2.2. Zero-Knowledge Proofs

Zero-knowledge proofs (ZKPs) are cryptographic protocols that allow a prover to convince a verifier that a statement is true without revealing any information beyond the validity of the statement [21]. Modern ZKP systems support verifiable computation over private inputs, so that complex predicates can be proven without disclosing the underlying data. In blockchain settings, ZKPs are widely used to reconcile public verifiability with privacy: privacy-focused systems such as Zerocash and Tornado Cash apply them to conceal transaction details or provide transaction anonymity [9,22]. These properties make ZKPs a natural building block for proving that a requester satisfies a hidden consent policy, which is the direction our approach develops.

2.3. The Data Use Ontology

The Data Use Ontology (DUO) provides a standardized vocabulary for expressing biomedical data-sharing consent conditions, and its terms are widely adopted in genomics and biobanking to enable semantic interoperability [6,16]. A DUO-based policy combines permissions and modifiers—for example, general research use (GRU), disease-specific research (DS), or a requirement that ethics-committee approval be obtained (IRB)—into the condition under which access is granted.

DUO Term Categories

Although DUO offers a rich set of permissions and modifiers, its terms are heterogeneous in how they must be evaluated for compliance. To structure their handling in a zero-knowledge setting, we group DUO terms into four categories:

- **Boolean-only:** simple yes/no conditions that can be represented directly without external validation, e.g. “no restriction” (NRES), general research use (GRU), or “health/medical/biomedical research” (HMB).
- **Ontology-linked:** terms whose evaluation involves consistency checks against external vocabularies or classifications, e.g. disease-specific research (DS), population origins or ancestry research (POA), or clinical care use (CC).
- **Obligation-required:** terms that impose procedural or temporal obligations, e.g. “publication required” (PUB), “publication moratorium” (MOR), “time limit on use” (TS), or “return to database” (RTN).
- **Credential-required:** terms whose evaluation relies on attested proof of requester status or approvals, e.g. “collaboration required” (COL), “ethics approval required” (IRB), “project-specific restriction” (PS), or “institution-specific restriction” (IS).

Table 1 summarizes representative DUO terms under this classification. Some terms reduce to simple Boolean flags, while others rely on ontology references, external obligations, or credential attestations. Section 5 describes how each category enters the proved relation and which requirements remain outside it (Table 4).

3. Related Work

Blockchain-based consent management for healthcare data sharing has been studied extensively, with smart contracts used to record, enforce, and audit consent across institutions [1–3]. This line of work focuses on making consent decisions verifiable and auditable, and the privacy mechanisms surveyed in this space mainly protect health data contents

Table 1. Representative DUO terms grouped by evaluation category.

Category	Example Terms	Description
Boolean-only	NRES (DUO:0000004), GRU (DUO:0000042), HMB (DUO:0000006)	Direct yes/no conditions, no external validation required.
Ontology-linked	DS (DUO:0000007), POA (DUO:0000011), CC (DUO:0000043)	Evaluated against external ontologies such as the International Classification of Diseases (ICD-10) or the Mondo Disease Ontology (MONDO).
Obligation-required	PUB (DUO:0000019), MOR (DUO:0000024), TS (DUO:0000025), RTN (DUO:0000029)	Impose obligations such as publication duties, time limits, or data return.
Credential-required	COL (DUO:0000020), IRB (DUO:0000021), PS (DUO:0000027), IS (DUO:0000028)	Depend on credentials, approvals, or institutional/project-specific restrictions.

and user identities [5]. The confidentiality of the governing consent policy—and what public-chain enforcement artifacts reveal about it—is typically out of scope.

Beyond the consent-management setting, existing work addresses important fragments of policy-private consent enforcement, but not their combination under public-chain observability. Encrypted data-sharing mechanisms, such as Ciphertext-Policy Attribute-Based Encryption (CP-ABE) and proxy re-encryption (PRE) [17,18], control access to encrypted data. However, standard CP-ABE attaches the access policy to the ciphertext in the clear, exposing the policy itself, while PRE-based delegation may reveal authorization patterns that data encryption is not designed to hide. Transaction-privacy systems, including Zerocash-style protocols [22] and Ethereum-oriented systems such as ZETH and AZTEC [7,8], provide public verification over private transaction witnesses, but they mainly target asset ownership, values, or participant relationships rather than compliance with a provider-defined consent policy.

Hidden-policy access-control work reduces direct policy exposure: CP-ABE variants hide the access structure in the ciphertext [13,14]. Wu et al. [23] instead realize attribute-based access control (ABAC) by encrypting policy and attribute values under an additively homomorphic cryptosystem. A smart contract computes their encrypted difference, one assistant node per organization decrypts it partially with a zero-knowledge proof of correct decryption, and the contract decides without seeing the values; every request therefore needs every organization online. Functional encryption gives a more general cryptographic abstraction for revealing function outputs while hiding inputs [24], but it does not by itself yield publicly verifiable authorization evidence. Private-state blockchain systems, including Secret Network and Oasis/Ekiden, protect smart-contract state or execution through encrypted state and trusted execution environments (TEEs) [25,26]. These lines of work address adjacent forms of policy or state hiding, but TPS-ZKPI targets a more specific setting: hiding consent policies on a public chain while retaining public verifiability of each authorization decision. Even where policies are hidden, existing designs may still leak policy-dependent artifacts through ciphertext structures, policy-update transactions, authorization workflows, or repeated interactions. In the homomorphic design, for example, the Boolean structure of the policy and its attribute names stay on the ledger, and the per-attribute match results can be recomputed from the ciphertexts and partial

decryptions recorded with each decision [23]. Moreover, many target permissioned or off-chain settings, such as that design's coalition blockchain, where enforcement artifacts are not globally visible as they are on public blockchains.

Self-sovereign-identity (SSI) architectures place verifiable claims under the requester's control and raise the converse question of which verifiers the holder can trust with those claims. Buccafurri et al. [27] answer it with ciphertext-policy attribute-based proxy re-encryption: the holder stores encrypted claims off-chain under a self-defined attribute policy, a semi-trusted, non-colluding consensus provider re-encrypts them for verifiers whose certified attributes satisfy that policy, and a smart contract notarizes hashes of storage, policy-change, and access operations so that they can later be attributed to their parties; a trusted key-generation authority issues all attribute keys. Zero-knowledge credential presentations likewise let a patient authenticate anonymously on public or private chains [28], and surveys of zero-knowledge proofs for blockchain privacy catalogue this design space [12,29]. TPS-ZKPI sits on the other side of the same authorization relation: the protected object is the provider's consent policy rather than the requester's claims, and the claims enter only as private witnesses of a proof with a fixed statement. Two differences follow. First, the smart contract decides rather than notarizes: it verifies the proof and records a context-bound outcome that any observer can check immediately, whereas notarization supports attribution after the fact. Second, off-chain trust is concentrated rather than split: their proxy never sees claim contents, whereas the TPS sees both the policy and the claims, which is why Section 4 treats the TPS as an explicit trust boundary and Table 3 states what survives its compromise. The designs are complementary: the requester could release its claims only to a TPS whose certified attributes satisfy a requester-defined policy, which is the verifier-authorization problem of [27] with the TPS as verifier, and a per-request holder signature over the authorization context would further prevent a TPS from reusing observed evidence for fresh sessions (Section 7).

Privacy-preserving access-control and policy-evaluation mechanisms have been studied in credential-based settings built on the eXtensible Access Control Markup Language (XACML) and the Security Assertion Markup Language (SAML) [30], as well as in policy evaluation by multi-party computation (MPC) using homomorphic encryption (HE) and secure function evaluation [31]. These approaches support confidential authorization computation, but they are not primarily designed for settings where policy-evaluation artifacts—calldata, public inputs, emitted events, timing, and gas usage—are permanently and globally visible on a public blockchain. In addition, these approaches do not directly provide compact evidence that a smart contract can verify against a committed, provider-defined policy instance.

Reusable proving systems, including universal-setup succinct non-interactive arguments of knowledge (SNARKs), composable proof systems, and zkVMs [32–34], improve maintainability by reducing the need for policy-specific circuits or verifier contracts. However, when deployed for public-chain consent enforcement, reusable proving alone does not guarantee policy privacy: policy-dependent program representations, public inputs, calldata layouts, logs, or gas profiles may still remain observable. This motivates TPS-ZKPI's use of a domain-specific reusable policy interpreter, which preserves proof reuse while exposing only a policy-independent public verification interface.

Public-chain metadata-leakage studies show that cryptographic privacy guarantees may be weakened by observable transaction-level artifacts such as timing and volume patterns [4]. Techniques such as batching, decoys, and traffic shaping can reduce some observable signals [35], but they are usually designed for payment privacy, anonymity sets, or transaction-linkability settings. They do not directly provide hidden-policy compliance verification or auditable consent enforcement. In contrast, TPS-ZKPI treats metadata

leakage as part of the policy-privacy objective rather than as a separate transaction-level concern.

Existing approaches address verifiable consent recording, data confidentiality, transaction privacy, policy hiding, confidential policy evaluation, proof reuse, or metadata mitigation in isolation, whereas policy-private consent enforcement under public-chain observability requires the combination of policy confidentiality, metadata-leakage control, verifiable auditability, and a stable verification interface for evolving Boolean-based consent policies. Table 2 summarizes this positioning.

Table 2. Positioning of representative approaches against policy-private consent enforcement under public-chain observability (Yes / Partial / No; assumptions in parentheses are part of the claim).

Approach	Policy confidentiality	Metadata-leakage mitigation	Verifiable auditability	Main limitation
Blockchain consent management [1,2]	No	No	Yes	Records and enforces consent verifiably on chain, but policy confidentiality and policy-dependent leakage are typically out of scope.
Data / transaction privacy [7,8,17,18,22,36]	No	Partial	Partial	Protects encrypted data access or transaction semantics, but not the governing consent policy or policy-dependent authorization workflow.
Hidden-policy ABE [13,14]	Partial	Partial	Partial	Reduces direct policy exposure, but may leak through workflows, updates, or policy-holder interactions.
Homomorphic hidden-policy ABAC [23]	Partial	No	Yes	Hides policy and attribute values on a permissioned ledger, but the Boolean structure, attribute names, and per-attribute match results remain recoverable from it.
SSI verifier authorization [27]	No	Partial	Partial	Releases the holder's claims only to attribute-qualified verifiers and notarizes access for later attribution, but protects the requester's claims rather than the provider's consent policy.
MPC / HE policy evaluation [30,31]	Yes	No	Partial	Supports confidential computation, but usually lacks public-chain-verifiable authorization evidence.
Universal SNARKs / zkVMs [32–34]	No	Partial	Yes	Supports proof reuse and public verification, but does not by itself hide policy-dependent public artifacts.
Metadata-leakage defenses [35]	No	Yes	No	Models or mitigates observable signals, but does not provide hidden-policy consent enforcement.
TPS-ZKPI	Yes (under TPS trust)	Yes (modeled channels)	Yes	Provides policy-private, metadata-aware, verifiable consent enforcement under the stated TPS trust assumption.

4. Problem Formulation

4.1. System and Trust Model

We consider a blockchain-based healthcare data-sharing system with two user roles: *data providers*, who provide protected resources and govern access through consent policies, and *data requesters*, who seek access to those resources subject to the applicable policies. TPS-ZKPI mediates this interaction through three components: a TPS operating off-chain, an on-chain access-control manager (ACM), and a data access layer. The provider defines a consent policy P for a protected resource on behalf of the data subject whose consent it encodes, and registers an on-chain commitment h to the encoded policy. The requester seeks access by submitting policy-relevant authorization evidence, such as attributes, credentials, and access context. The TPS constructs the private witness and generates a zero-knowledge proof of policy compliance. The ACM verifies the proof on-chain

and records the authorization outcome, while the data access layer grants access according to the recorded decision.

TPS-ZKPI relies on the following assumptions. First, policies are provider-controlled: the provider is responsible for defining the applicable consent policy and committing the corresponding policy instance on-chain. Second, the TPS is trusted to preserve the confidentiality of policy-bound proving material and requester-submitted authorization evidence, and to follow the prescribed proof-generation workflow. Third, the commitment scheme is assumed to be hiding and binding, and the zero-knowledge proof system is assumed to satisfy completeness, soundness, and zero knowledge with respect to the implemented policy-compliance relation. Issuer governance, credential issuance and revocation, and real-world identity verification are treated as external lifecycle functions.

Conceptually, the TPS is an independent proof service, distinct from the provider; this separation is what makes relocating trust meaningful, since it prevents the provider from being the single party that holds both the policy and the requester evidence. Possible instantiations of such an independent service include a third-party proof service, a TEE-backed proof service that makes the trust assumption hardware-attested, or MPC/-collaborative SNARK proof generation in which no single party sees both policy material and requester evidence. For evaluation simplicity, the current evaluation co-locates the TPS with the provider (a provider-operated TPS). This co-location concentrates trust at the provider side: a single party now holds both the policy and the requester evidence, so a compromised provider-operated TPS could substitute inputs whose origin is not independently authenticated or withhold proof generation from legitimate requesters.

Two guarantees must be distinguished here. Authorization correctness is primarily enforced by the commitment and on-chain proof verification: a proof that does not satisfy the committed policy relation is rejected, largely independently of how the TPS is instantiated. Policy confidentiality, in contrast, is not cryptographically enforced against the TPS itself; it rests entirely on the trust assumption stated above. Our evaluation therefore assumes a TPS that follows the prescribed workflow and isolates the remaining question: what a public-chain observer can infer from the resulting public artifacts. Weakening the TPS trust assumption itself is orthogonal and is the role of the reduced-trust instantiations above. Table 3 summarizes, for each actor or failure mode, which of the four guarantees—policy confidentiality, authorization correctness, liveness, and decision auditability—is enforced cryptographically, which holds only under the TPS trust assumption, and which the prototype leaves to external functions or does not provide.

Attribute-origin guarantees depend on the evidence type. For DUO terms evaluated from the claims in an issuer-signed profile credential presented by the requester (hereafter, profile-backed terms), the prototype verifies the EdDSA-Poseidon credential under an issuer key selected by the provider and bound to the policy commitment. EdDSA-Poseidon denotes the Edwards-curve Digital Signature Algorithm (EdDSA) [37] over the Baby Jubjub curve [38], instantiated with Poseidon [39], a hash function designed for efficient evaluation inside zero-knowledge circuits. The credential binds the policy commitment, requester pseudonym, protected resource, claims, validity interval, and credential identifier; the corresponding policy attributes are derived inside the circuit. Thus, subject to signature unforgeability and Poseidon collision resistance, these values cannot be modified without invalidating the proof. This guarantee does not establish the real-world authority of the issuer or the truthfulness of issuance. Ontology and requester-allowlist predicates likewise establish membership only relative to provider-committed sets. Issuer-key governance, credential revocation, real-world identity verification, and protection against a compromised TPS remain external trust and lifecycle functions.

Table 3. Guarantees under each actor or failure mode.

Actor or failure mode	Policy confidentiality	Authorization correctness	Liveness	Decision auditability
Passive public-chain observer	C for proof and commitment contents; empirical for modeled metadata	C	n/a	C: outcome recorded
Untrusted requester	C against disclosure from public artifacts; adaptive-query leakage not evaluated	C: cannot forge a proof or alter its bound authorization context	n/a	C: pseudonymous request context and outcome recorded on-chain
Workflow-following TPS	– against the TPS itself; T against onward disclosure	C for the committed relation; T/E for unsigned evidence provenance	T	C for submitted decisions; – for refusal
Compromised or colluding TPS	–	C for the committed relation and signed claims; E for externally authenticated evidence; – for unauthenticated evidence provenance	–: may refuse to prove	C for submitted decisions; – for refusal
Forged credential	n/a	C: signature verified under the policy-bound issuer key	n/a	n/a
Unauthorized or revoked issuer	n/a	E: issuer authorization and revocation status are external	n/a	C for the on-chain decision record; E for determining issuer status

Note: C: cryptographic; T: trust in the TPS following the prescribed workflow; E: external or lifecycle function; –: not provided by the prototype; n/a: not applicable. In the evaluated provider-operated deployment, the provider and TPS are co-located, so all TPS-dependent confidentiality and liveness assumptions are concentrated at the provider.

4.2. Public-Chain Adversary

The primary adversary is a passive public-chain observer. The adversary can inspect smart-contract code and monitor public blockchain artifacts across multiple authorization sessions, including policy commitments, submitted proofs, public inputs, emitted events, calldata, timing, and gas usage. The adversary does not observe provider-to-TPS policy material, commitment-opening randomness, requester-submitted authorization evidence, credential witnesses, or TPS-side witness construction: these items are exchanged over off-chain channels or held locally by the provider and the TPS, and never appear in on-chain artifacts. We assume the off-chain channels themselves are confidential (e.g., TLS-protected).

Each policy-enforcement instance corresponds to an authorization session, in which a requester seeks access to a protected resource under a committed policy. For each such session, the adversary observes a public view

$$view_{\mathsf{pub}} = (h, \pi, pp, d_\chi, \tau, id_{\mathrm{req}}, id_{\mathrm{res}}, \nu, meta), \tag{1}$$

where h is the on-chain commitment to the hidden policy, π is the zero-knowledge proof submitted for on-chain verification, pp, d_χ, and τ are the remaining public signals of the

proof statement (a parameter commitment, a context digest, and a request epoch; Section 5), id_{req}, id_{res}, and ν are the requester, resource, and session-nonce values carried in the access transaction, and *meta* denotes transaction-level metadata exposed by the enforcement layer, such as calldata footprint, event structure, public-input layout, timing, and gas usage. The adversary's goal is to infer hidden information about the committed policy or the authorization interaction from this public view.

In addition to the public-chain observer, we treat the requester as untrusted with respect to policy confidentiality. The requester initiates access requests, observes its own submitted evidence and authorization outcome, and inspects the public transcript. However, the requester does not receive the encoded policy, commitment-opening randomness, or policy-bound proving material, which remain within the provider-to-TPS proof-generation boundary. This restriction matters because a requester that learned the policy could infer sensitive facts about the data subject from the consent terms themselves and, in addition, would learn the policy's full acceptance region—which attribute combinations are granted—enabling targeted evidence construction or probing beyond its own legitimate sessions.

Adaptive requester probing and off-chain side channels are outside the scope of the present evaluation. The proved authorization outcome is bound to a pseudonymous requester identifier, a protected resource, and a session nonce through the context digest defined in Section 5. This binding prevents substitution of the recorded authorization context, but it should not be interpreted as proof of the requester's real-world identity.

4.3. Privacy Objectives

We operationalize policy privacy as three measurable, attack-oriented objectives that cover the information a public-chain observer could extract about a hidden policy: its semantic content, its identity relative to other policies, and its structural form. These correspond to Policy Content Hiding, Policy Indistinguishability, and Structure and Parameter Privacy, respectively.

Policy Content Hiding (PCH). PCH requires that public-chain artifacts should not reveal the semantic content of the hidden consent policy, such as data-use terms, disease categories, research purposes, institutional restrictions, or ethics-related requirements.

Policy Indistinguishability (PI). PI requires that different candidate policies should not become distinguishable from their public verification artifacts when they are compiled into the same fixed public interface. This captures whether two policies can be told apart from commitments, proofs, public parameters, or metadata even when their direct representations are hidden—that is, whether the identity of the governing policy can be narrowed down.

Structure and Parameter Privacy (SPP). SPP requires that public-chain artifacts should not reveal structural or parametric properties of the hidden policy, such as logical structure, nesting depth, predicate count, set size, or threshold-like parameters.

4.4. Problem Statement

Given a provider-defined consent policy P, requester authorization evidence, a globally observable public blockchain, and the TPS trust assumption of Section 4.1, the problem is to enforce access to a protected healthcare resource subject to four requirements. First, authorization must be publicly verifiable and bound: the ACM, and hence any observer, can check that the requester satisfies the committed policy without learning or executing it in plaintext, and an accepting proof is tied to the complete committed policy and to the authorization context of the request (requester identifier, resource, and session), so that it cannot be reused for another context. Second, the policy must remain confidential:

public-chain artifacts should not reveal its content, identity, or structure and parameters (PCH, PI, and SPP; Section 4.3), except for terms whose enforcement the design deliberately makes public (Section 5); requesters should learn nothing about it beyond their own authorization outcomes; and requester evidence should not appear on-chain. Third, authorization decisions should be auditable: each processed request, granted or denied, is recorded on-chain together with its bound context and can be checked later without exposing the policy logic. Fourth, one policy-independent verification interface should serve every policy in the supported class, so that policies can be added or changed without changing the on-chain verifier. The first, third, and fourth requirements are met by construction (Section 5); the second is evaluated empirically against the modeled public-chain adversary (Section 6).

5. The Proposed Approach: TPS-ZKPI

5.1. Overview

As shown in Figure 1, the TPS-ZKPI workflow binds a provider-committed hidden policy to requester-submitted authorization evidence through TPS-side proof generation and on-chain proof verification. The provider registers only a policy commitment on-chain and sends policy-bound proving material to the TPS off-chain (steps 1–2 in the figure). For each access request, the TPS constructs the private witness from the hidden policy material and requester evidence, generates a zero-knowledge proof of policy compliance, and submits a fixed-format proof statement for ACM verification (steps 3–4). The ACM records the authorization outcome in contract storage and emits it in the blockchain event log (as access-granted or access-denied events; step 5); the data access layer consults the recorded decision to serve or refuse the request (steps 6–9).

This design separates confidentiality-sensitive proof generation from public verification. Policy material and requester evidence remain off-chain—policy material within the provider–TPS boundary, requester evidence between the requester and the TPS—while the blockchain verifies only a fixed-format proof statement against the provider's policy commitment, alongside session parameters validated by the ACM. In a traditional requester-as-prover design, the requester generates the proof itself and must therefore receive the encoded policy and the commitment-opening randomness, so every requester learns the policy in normal operation. In TPS-ZKPI, this material never leaves the provider–TPS boundary: the requester receives no policy material, and the policy itself is disclosed only if the TPS violates its trust assumption; what the public transcript reveals about it is evaluated in Section 6.

5.2. Policy Representation and Interpretation

TPS-ZKPI represents heterogeneous Boolean-based consent policies in a fixed-length, circuit-friendly format. This policy class is chosen for two reasons. First, the supported biomedical consent conditions—DUO-style data-use terms, disease-scope constraints, and requester restrictions—are categorical conditions combined with AND/OR, and conditions that are not natively Boolean enter through fixed external-predicate slots. The generic interpreter also evaluates NOT, but the DUO compiler does not expose negation because its membership witnesses establish inclusion rather than non-membership. Second, a Boolean policy class admits a fixed-length encoding evaluated by a single reusable interpreter circuit, which keeps the verification interface policy-independent—a prerequisite for the privacy objectives evaluated later. Let

$$X = \{0,1\}^N, \qquad B = \{0,1\}^M \tag{2}$$

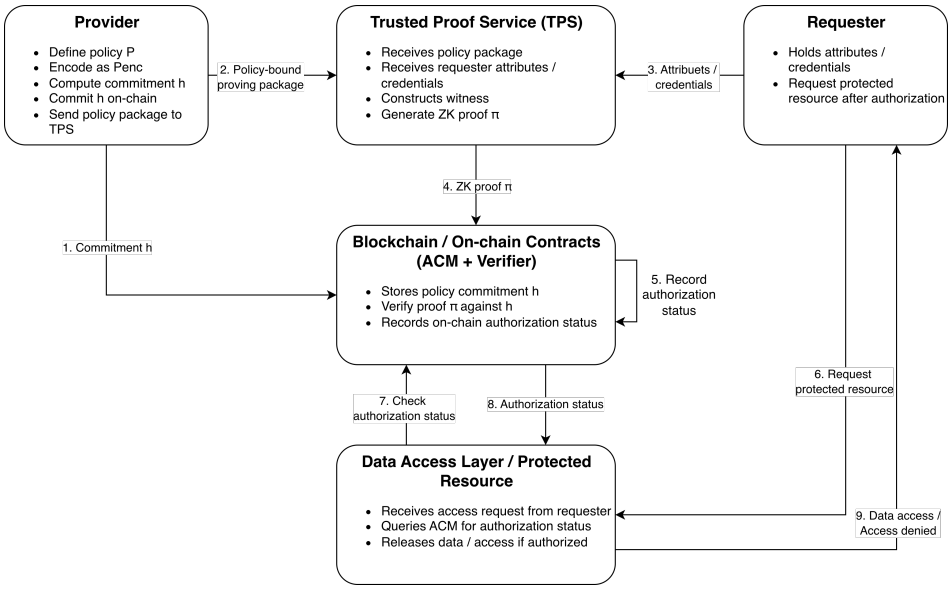


Figure 1. An overview of the TPS-ZKPI workflow.

denote the requester-attribute assignment space and the external-predicate assignment space, respectively. The full assignment space is

$$O = X \times B, \tag{3}$$

where $a \in X$ encodes requester attributes and $b \in B$ encodes predicate outcomes, including credential, ontology-membership, requester-allowlist, and obligation-related checks. The implemented relation derives these bits from their associated witnesses rather than accepting unconstrained predicate assertions. The strength of the resulting provenance guarantee nevertheless depends on the mechanism: the profile-credential slot verifies an issuer signature in-circuit, whereas ontology and requester-allowlist slots establish membership relative to provider-committed sets. Issuer authorization, issuance practice, and credential revocation remain external lifecycle functions (Section 4.1).

Ontology-linked conditions are represented extensionally rather than through in-circuit ontology reasoning. At encoding time, the provider expands an ontology concept into a whitelist of codes and commits the whitelist as a Merkle root among the policy's per-predicate parameters; in-circuit, the predicate check is a set-membership proof of the requested term against this root. A disease-specific policy scoped to diabetes, for example, whitelists the diabetes subtree of the International Classification of Diseases, 10th Revision (ICD-10), whereas one scoped to type-1 diabetes whitelists only the type-1 codes. Resolving subsumption at encoding time keeps ontology reasoning out of the circuit and preserves the policy-independent interface.

Consent policies are expressed as Boolean formulas over assignment bits:

$$P ::= atom \mid (P \wedge P) \mid (P \vee P) \mid \neg P, \qquad atom ::= x_i \mid y_j. \tag{4}$$

Here x_i selects requester-attribute bit a_i and y_j selects external-predicate bit b_j. For example, the consent condition "the resource may be used for disease-specific research on diabetes, or for clinical care under approved clinical terms" is expressed as $(x_1 \wedge y_1) \vee (x_2 \wedge y_2)$, where x_1 and x_2 flag the disease-specific-research and clinical-care purposes among the requester attributes, y_1 attests that the requested disease code lies in the committed diabetes whitelist, and y_2 attests that the requested clinical term lies in the approved clinical whitelist. Each policy is compiled into a Reverse Polish Notation (RPN) program over

operand tokens and Boolean operator tokens. RPN linearizes arbitrarily nested formulas into a parenthesis-free token sequence that a stack machine evaluates in a single left-to-right pass with a fixed step count—a control structure that maps directly onto a fixed-size circuit and lets one interpreter serve all policies. To support reusable verification, each program is padded to a fixed token length L:

$$P_{\text{enc}} = [t_1, \ldots, t_L], \tag{5}$$

together with an explicit program length $\ell \leq L$ carried as a private witness. The interpreter evaluates only the first ℓ tokens; padding positions beyond ℓ are inactive and leave the output unchanged. The example policy above compiles to the token sequence $[x_1, y_1, \wedge, x_2, y_2, \wedge, \vee]$ with $\ell = 7$; the remaining $L - 7$ positions are padding. The implemented circuit uses $N = 13$, $M = 11$, and $L = \mathsf{DEPTH} = 64$. We denote the set of length-L programs by Prog_L.

The reusable policy interpreter is a zero-knowledge circuit

$$C : \mathsf{Prog}_L \times \{0, \ldots, L\} \times O \to \{0, 1\}. \tag{6}$$

Given an encoded policy, its program length, and operand values, the interpreter evaluates the RPN program using deterministic stack semantics and enforces well-formedness constraints inside the circuit. The circuit outputs 1 only if the requester evidence satisfies the encoded policy. Well-formedness is enforced as hard constraints: for a program that violates the stack discipline, no satisfying witness exists and no proof can be generated. An output of 0 therefore indicates unsatisfied evidence for a well-formed program. The proved acceptance relation is the asserted case $C(P_{\text{enc}}, \ell, (a, b)) = 1$.

The fixed-length interpreter thus realizes the policy-independent interface motivated above, and supports policy evolution through different fixed-length policy programs.

Scope of the supported policies. The policy class is restricted, and Table 4 states what the prototype establishes end-to-end. Unrestricted use (NRES) is supported only as a standalone policy; other supported terms may be combined by AND/OR in programs of at most L tokens, with each term appearing at most once and MOR/TS restricted to conjunctions. Categorical data-use terms, ontology-scoped conditions, requester allowlists, profile-backed terms, the policy validity window, and time-limited use are decided at authorization time, either inside the proved relation or by the ACM from public state; profile-backed terms establish possession of issuer-signed claims, not conformity of subsequent data use with a declared purpose. Because the ACM enforces time-limited use from public state, a policy containing TS is not content-private for that term: the commit transaction emits the duration, and the first grant emits the requester's window. The evaluated corpus contains no TS policies (Section 6.3). A publication moratorium is bound into the policy as an accepted obligation but cannot be decided at request time, and institutional or ethics approval, downstream purpose limitation, post-access duties, delegation, and credential revocation are left to external governance and the issuer lifecycle. On-chain policy revocation and resource rebinding affect future authorization decisions only; they do not retract data already released or enforce cessation or deletion after access. These lifecycle transactions are public: they reveal when a policy is replaced or revoked, but not its content. Accordingly, the prototype is an access-time consent checker for bounded Boolean policies over standardized terms.

Table 4. Consent-policy features and how far the prototype establishes them. "In-circuit" means the condition is part of the proved relation; "on-chain" means the ACM enforces it from public state; "committed" means the value is bound into the policy but compliance is not decided at request time.

Consent feature	Status	What is established at authorization time
Unrestricted use (NRES)	In-circuit	Compiled as a standalone unconditional policy; it cannot be combined with another DUO term.
Categorical data-use terms combined with AND/OR	In-circuit	The committed program and its length are evaluated by the interpreter; each term may appear at most once.
Disease, study-design, ancestry, geography, and clinical-term scopes (DS, RS, POA, GS, CC)	In-circuit	Merkle membership of the requested code in a provider-committed whitelist whose root is bound in the parameter commitment.
User-, project-, and institution-specific restrictions (US, PS, IS)	In-circuit	Membership of the holder pseudonym in a provider-committed allowlist for the scope.
Profile-backed terms (GRU, HMB, NMDS, GSO, NPUNCU, NPOA, NPU, NCU)	In-circuit	EdDSA-Poseidon credential over policy commitment, pseudonym, resource, claims, validity interval, and identifier; the attributes are derived from the signed claims, but subsequent conformity with a declared purpose is not established.
Policy validity window	In-circuit + on-chain	The committed window is checked against the public request epoch, which the ACM requires to equal the current hour.
Time-limited use (TS)	On-chain	The committed duration is also stored and emitted in clear by the ACM; it starts a per-requester window on the first grant, and later requests outside it are denied. TS may appear only conjunctively.
Publication moratorium (MOR)	Committed	The deadline is bound into the policy as an accepted obligation; publication conduct is not observable at request time. MOR may appear only conjunctively.
Prospective consent withdrawal and policy evolution	On-chain	A successor policy supersedes its predecessor, a policy can be revoked, and a resource can be rebound for future decisions; this does not retract data already released or enforce downstream cessation or deletion.
Institutional or ethics approval (IRB)	Not provided	Documentary approval and committee review remain an out-of-circuit authorization step.
Downstream purpose limitation, post-access duties (PUB, COL, RTN), delegation, credential revocation	Not provided	Carried by external governance, the data access agreement, and the issuer lifecycle, outside the proved statement.

5.3. Commitment Binding and Proof Generation

To bind each proof to a provider-defined hidden policy, the provider samples commitment randomness r uniformly from the scalar field

of BN254, the pairing-friendly elliptic curve over which the proofs are computed, and computes

$$\begin{aligned} q &= \mathsf{H}_{\mathrm{pol}}(P_{\mathrm{enc}}), \\ h &= \mathsf{Com}(P_{\mathrm{enc}}, \ell; r) = \mathsf{Poseidon}(q \parallel \ell \parallel r), \end{aligned} \tag{7}$$

where q is the Poseidon-based digest of the fixed-length encoded policy, ℓ is the private program length, r is the commitment randomness, and h is the policy commitment stored on-chain. The semantics of an encoded policy are determined by the pair (P_{enc}, ℓ), since ℓ identifies the active prefix of the padded token array. Accordingly, the circuit supplies the same private value ℓ to both the commitment verifier and the policy interpreter. Under the collision resistance of Poseidon, an accepting proof bound to h cannot substitute a different

program length and therefore cannot evaluate an alternative stack-consistent prefix of P_{enc}. Hiding is an explicit commitment assumption (Section 4.1) that relies on high-entropy r and the fixed-length encoding of P_{enc}, which keeps the sponge input unambiguous.

For an access request, let

$$\chi = (id_{\text{req}}, id_{\text{res}}, \nu), \qquad d_\chi = \mathsf{Poseidon}(id_{\text{req}} \parallel id_{\text{res}} \parallel \nu), \tag{8}$$

where ν is a fresh session nonce. Here id_{req} is a pseudonymous authorization identifier. For policies that activate a holder-bound credential or requester-allowlist predicate, the circuit derives it as $\mathsf{Poseidon}(s_{\text{holder}})$ from a private holder secret s_{holder}; otherwise, it may be a requester-supplied pseudonym. The TPS constructs the private

circuit witness

$$w = (P_{\text{enc}}, \ell, a, b, r, \omega, \chi), \tag{9}$$

where P_{enc} is the encoded policy, ℓ is its program length, a denotes requester attributes, b denotes external predicate outcomes, r is the commitment randomness, and ω denotes auxiliary witnesses for credential- or predicate-dependent checks. The context tuple χ consists of private circuit inputs, while its digest d_χ is public. The current public-signal vector is

$$(1, h, pp, d_\chi, \tau), \tag{10}$$

where pp is a fixed-format commitment to the per-predicate parameters and τ is the request epoch used by the temporal checks. Although the three components of χ are not separate Groth16 public inputs, the ACM receives them as transaction parameters in order to recompute d_χ; they are therefore visible in transaction calldata.

For the compiled circuit, the Groth16 [21] setup produces the common reference string as a proving key and a verification key, once for the shared circuit:

$$(pk, vk) \leftarrow \mathsf{Setup}(C). \tag{11}$$

The TPS generates a proof

$$\pi \leftarrow \mathsf{Prove}\big(pk, w; (1, h, pp, d_\chi, \tau)\big). \tag{12}$$

The proof certifies the conjunction that $C(P_{\text{enc}}, \ell, (a, b)) = 1$, $h = \mathsf{Com}(P_{\text{enc}}, \ell; r)$, $d_\chi = \mathsf{Poseidon}(\chi)$, and the selected external-predicate witnesses satisfy their policy-bound verification relations. Before verifying π, the ACM checks that the resource is governed by the stated policy, that τ is current, that the nonce has not been used for the same requester–resource context, and that the digest recomputed from the transaction parameters equals the public signal d_χ. Thus, within one ACM deployment, a proof generated for one requester identifier, resource, and nonce cannot be accepted under a different tuple. The ACM then verifies π against vk and records the outcome of each request: a positive authorization outcome requires both a valid proof and an asserted output of 1; otherwise the request is recorded as a denial. The policy encoding, commitment randomness, requester attributes, credential witnesses, and auxiliary proof material are not posted on-chain.

5.4. Credential-Origin Binding

The circuit reserves one fixed predicate slot for a credential carrying a fixed set of DUO-relevant claims. Let $c_0, \ldots, c_6$ be the private, issuer-signed claim bits for, respec-

tively, research use, health or medical use, methods development, genetic study, ancestry, commercial use, and non-profit use. Let t_{from} and t_{until} define its validity interval, and let γ be its credential identifier. Let d_{DUO} be the fixed credential schema/domain tag for this credential type. From these signed claims, the circuit derives the DUO terms GRU, HMB, no general methods research (NMDS), genetic studies only (GSO), not-for-profit, non-commercial use only (NPUNCU), population origins or ancestry research prohibited (NPOA), not-for-profit use only (NPU), and non-commercial use only (NCU). For issuer public key $A = (A_x, A_y)$, the circuit reconstructs the domain-separated message

$$m_{\text{cred}} = \mathsf{Poseidon}(d_{\text{DUO}} \parallel h \parallel id_{\text{req}} \parallel id_{\text{res}} \parallel c_0 \parallel \cdots \parallel c_6 \parallel t_{\text{from}} \parallel t_{\text{until}} \parallel \gamma) \tag{13}$$

Before proof generation, an issuer holding the corresponding private signing key s_{iss} issues the credential by computing

$$\sigma \leftarrow \mathsf{Sign}_{\text{EdDSA-Poseidon}}(s_{\text{iss}}, m_{\text{cred}}). \tag{14}$$

The resulting credential is supplied to the TPS as private off-chain evidence for proof generation. At proof generation, the TPS supplies A, σ, and the credential fields as private witness values. The circuit then enforces

$$\mathsf{Verify}_{\text{EdDSA-Poseidon}}(A, \sigma, m_{\text{cred}}) = 1, \qquad \mathsf{Poseidon}(A_x \parallel A_y) = k_{\text{iss}}, \tag{15}$$

where k_{iss} is the issuer-key commitment contained in the policy-bound parameter commitment pp. The circuit additionally requires $t_{\text{from}} \leq \tau \leq t_{\text{until}}$ and $\gamma \neq 0$. Because this credential is holder-bound, a policy that activates this slot also requires $id_{\text{req}} = \mathsf{Poseidon}(s_{\text{holder}})$, binding the credential to the pseudonymous requester identifier in the authorization context. The corresponding DUO attributes are constrained to these claim-derived values.

5.5. Transcript Normalization

Zero-knowledge proofs hide private witness values, but surrounding blockchain artifacts may still leak policy-dependent information. TPS-ZKPI therefore normalizes the public transcript so that different policies induce fixed or tightly bounded observable profiles. Table 5 summarizes the mechanisms.

Table 5. Transcript-normalization mechanisms in TPS-ZKPI.

Artifact	Normalization mechanism
Calldata	Fixed verification interface and fixed-size proof/public-input encoding.
Logs	Fixed event structure with constant topic and payload layout.
Public parameters	Fixed-format handles instead of policy artifacts or credential witnesses.
Gas usage	Policy-independent verifier logic and gas padding toward a target drawn at random from a fixed band.

These mechanisms do not eliminate all public observability. Rather, they reduce policy-dependent variation in calldata, logs, public parameters, and gas-related behavior that could otherwise serve as a fingerprint for the metadata-only adversary defined in Section 4. TPS-ZKPI does not flatten all residual channels: timing between transactions, transaction ordering, mempool or broadcast patterns, and off-chain TPS communication remain outside the normalized public transcript.

6. Experiments and Results

We evaluate PCH, PI, and SPP (Section 4.3) against a passive public-chain observer (Section 6.1). The evidence comes from two attack families run on the same executions:

a supervised attacker that tries to recover each hidden label from the on-chain observables, and an unsupervised attacker that asks whether the observables cluster along that label without seeing it. Across the 45 tasks, two meet the failure criterion (one ICD-10 chapter and one nesting-depth contrast); both trace to the random pad target, and the unsupervised attacker finds no natural grouping by the hidden labels beyond the same coincidence. Table 6 summarizes the headline supervised verdicts across the three privacy objectives. Beyond the privacy verdicts, we report the system's performance as a cost profile: per-transaction gas for commit and access (including the padding overhead), calldata sizes, circuit constraint count, and proving time, together with how circuit capacity, fixed-circuit policy complexity, and offline proof throughput scale.

Table 6. Headline supervised results on the Polygon Amoy dataset (30 policy-disjoint seeds; a task fails iff the mean area under the receiver operating characteristic curve, AUC_{mean}, exceeds 0.55 and at least 50% of seeds are significant).

Property	Hidden target	Headline supervised result
PCH	DUO terms and ICD-10 chapters	19 of 20 content-recovery tasks pass; AUC 0.464–0.570, significant seeds median 3%. ICD ch5 fails (AUC 0.667, 53%); its separability is carried entirely by the random pad target*
PI	Related policy-pair identity	3 gating tasks pass; AUC 0.509–0.549; significant seeds median 7%, max 10%
SPP	Structure, predicate, parameter, and nesting-depth buckets	20 of 21 pairwise contrasts pass; AUC 0.468–0.571, significant seeds median 7%. `nestingDepth` medium/deep fails (AUC 0.563, 53%); carried by the pad target*

* Rerunning the same protocol on the gas minus the logged pad target returns both tasks to PASS (Sections 6.4 and 6.6).

6.1. Adversary and observables

We instantiate the passive public-chain observer of Section 4: an adversary who reads the ledger without restriction and sees every transaction and event our approach emits, but never the plaintext policy or the witness.

Under the assumptions of Section 4.1, the byte values in the adversary's public view carry no policy information: the proof is zero-knowledge, the commitments in the public statement are hiding, and the authorization output is identical for every execution in the corpus.

The attack therefore targets the metadata of these artifacts: for each policy execution (a commit transaction followed by an access transaction) it extracts the following features:

- **Gas:** `commitGasUsed`, `accessGasUsed`.
- **Calldata length:** `commitCalldataLength`, `accessCalldataLength`.
- **Proof size:** `proofSize`.
- **Event logs:** `commitLogsCount`, `commitLogsSize`, `accessLogsCount`, `accessLogsSize`.

Of these features, calldata length, proof size, and log counts/sizes are constant across the evaluated policies by design; any policy-dependent signal can therefore only flow through the two gas measurements (`commitGasUsed`, `accessGasUsed`). The fixed circuit and the commitment-only interface make the non-gas features constant; randomized gas padding protects the gas channel that remains. That channel is wider than the verifier's arithmetic: because the Ethereum Virtual Machine (EVM) prices calldata per byte (four gas for a zero

byte, sixteen otherwise), the gas of an access transaction also encodes the byte pattern of its public arguments, so any deterministic difference in how an argument is encoded reaches the observer through gas even when the calldata length is constant (Section 6.9 exhibits one such difference). The constant features remain in the attack pipeline but carry no variation to exploit. Time-limited use (TS) is the exception by design: the ACM enforces it from public state, so a TS policy discloses the term and its duration through additional events (Section 5); the evaluated corpus contains no TS policies. We do not model side channels such as block timing, transaction ordering, mempool or broadcast patterns, or off-chain TPS communication, nor the non-goals stated in Section 4.

We consider three adversary goals, one per privacy objective: (i) recover the policy's *content* (PCH, Section 6.4); (ii) distinguish two *semantically related* policies (PI, Section 6.5); and (iii) recover the policy's *structure or parameter sizes* (SPP, Section 6.6).

6.2. Attack methodology

We evaluate each privacy objective with a primary supervised attack and a complementary unsupervised cross-check. The *supervised* method asks whether a trained classifier can recover a hidden label from the observables, measured by the area under the receiver operating characteristic curve (AUC) against the chance level of 0.5. Its outcome serves as the main privacy verdict. The supervised adversary has strictly more information than the unsupervised one, and is thus the stronger of the two. Because this adversary is harder to defeat, a PASS under the supervised test is the more conservative verdict. The *unsupervised* method asks whether the observables cluster along the hidden label, measured by the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) against their chance level of 0. This chance level is exact for ARI but only nominal for NMI, which is positively biased at small samples (Section 6.8).

Execution setup. All policy executions are transactions on the *Polygon Amoy* testnet: for each generated policy, we commit the encoded policy on-chain from a provider account (`commitPolicy`), generate the Groth16 proof locally in the role of TPS (Section 4.1), and submit the access transaction from a separate TPS account (`requestAccessBound`). The provider-to-TPS and requester-to-TPS exchanges are in-process calls: they use no public channel and produce no on-chain artifacts. They therefore do not affect the observables of Section 6.1, which are extracted from the resulting transaction receipts, calldata, and event logs.

Both methods run on the same dataset of *Polygon Amoy* executions, collected under the full defense configuration (fixed circuit, commitment-only interface, randomized gas padding). The dataset, attack code, and verdict files are released as an accompanying artifact (see the Data Availability Statement).

6.2.1. Supervised attacks

Each objective is operationalized as a set of binary *tasks*, each defined by one hidden label to be recovered from the observables: detecting the presence of a specific data-use term or disease chapter (PCH, Section 6.4), distinguishing executions of two policies (PI, Section 6.5), or separating two structural buckets (SPP, Section 6.6). For every task we train a linear and a non-linear classifier (logistic regression, LR; random forest, RF) and report the score that enters the verdict for that objective (defined below). Each task is evaluated over **30 random seeds** using a **policy-disjoint train/test split** (the same 70/30 ratio in each of PCH, PI, and SPP): the policies used to train the attacker never appear in the test set, so a high test AUC reflects leakage that generalizes across policies. Because the split assigns whole policies rather than individual executions, we cannot partition executions to hit the target class proportions exactly. For each seed we therefore approximate stratification by

sampling many candidate splits (512 per seed in our runs) and keeping the one whose per-class test proportions are closest to 70/30. Here a class is one side of the binary task (the positive or negative value of its hidden label). By keeping each class's test share close to the target ratio, the candidate search ensures every seed's test set contains both classes, so each seed's AUC is estimated on a two-class test set.

For each seed we assess the AUC with a label-permutation test. The classifier is trained once and its test-set probability outputs are fixed; the test-set labels are then shuffled R=500 times, and the AUC is recomputed against the fixed probabilities on each shuffle, giving the null distribution of the AUC under no association between scores and labels. The empirical p-value is $(k+1)/(R+1)$, where k is the number of shuffles whose AUC meets or exceeds the observed (unshuffled) AUC; the added counts keep the p-value away from an exact zero. The seed is counted as significant when this p-value falls below α=0.05.

For each task, the result tables report the AUC averaged over seeds (AUC_{mean}) and the mean per-seed distance from chance, $\text{mean}_s\,|\text{AUC}_s - 0.5|$ (reported as the AUC gap). Because the gap is computed before averaging, deviations above and below 0.5 cannot cancel. Consequently, $\text{mean}_s\,|\text{AUC}_s - 0.5|$ is always at least as large as $|\text{AUC}_{\text{mean}} - 0.5|$.

Because every task is binary, its AUC has a chance level of 0.5 regardless of class balance. Thus, differences in positive and negative sample counts across the generated corpus do not shift the chance baseline, although they may affect estimation uncertainty.

The PASS/FAIL decision, defined below, combines AUC_{mean} with the significant-seed fraction—the fraction of seeds whose per-seed permutation test rejects at α=0.05, reported as %sig in the result tables. The verdict is:

> **FAIL** iff $\text{AUC}_{\text{mean}} > 0.55$ **and** the significant-seed fraction is ≥ 0.5 (over 30 policy-disjoint seeds); **PASS** otherwise.

Both conditions of this *double criterion* are required to trigger a FAIL verdict: the AUC threshold imposes a minimum effect size, while the significant-seed threshold requires the effect to recur across policy-disjoint splits. A task that does not satisfy both conditions is labeled PASS under this operational criterion; PASS does not constitute proof of zero leakage.

Operational thresholds and sensitivity. The PASS/FAIL labels summarize residual distinguishability under the evaluated attacks; they do not by themselves establish the effectiveness of the defense. The two conditions guard against different errors. The AUC threshold is an effect-size floor of 0.05 above chance: with large test sets, a permutation test can reject for a negligible AUC excess, and the floor keeps such an effect from triggering a FAIL. The stability condition requires the permutation-test p-value to be below 0.05 in at least 15 of the 30 policy-disjoint splits. Under the null hypothesis, each split rejects with probability 0.05, so a majority of rejections indicates an effect that recurs across splits rather than one carried by a few favorable splits. Because the splits overlap, the significant-seed fraction is a stability measure rather than a combined p-value. In the evaluated run, the stability condition is the binding one: the AUC condition alone would flag four PCH and four SPP tasks, three of each being significant in fewer than half of the splits, whereas the stability condition alone flags the same single task per experiment as the double criterion. The AUC floor therefore acts as a safeguard rather than as the deciding condition.

Because these thresholds are operational choices, we reapply the criterion over AUC thresholds of 0.52–0.60 and significant-seed fractions of 0.3–0.7. At fractions of at least 0.5, at most the two tasks discussed in the PCH and SPP result sections fail; lowering the fraction to 0.3–0.4 adds at most two SPP contrasts, whereas raising it to 0.6 yields none. No PI task fails under any of these threshold pairs. In ten null reruns with policy-independent gas features, four contain a spurious PCH failure (a 3% per-task frequency), while none

produces an SPP failure. We therefore treat FAIL as a diagnostic flag and examine the two observed failures in the corresponding result sections without changing their verdicts.

The effectiveness of gas normalization is assessed separately by the paired gas-normalization ablation. Using the same policies, features, splits, and classifiers, the attacks recover signal from the unpadded transcript, and that signal collapses after padding. Thus, the PASS/FAIL criterion summarizes residual attack performance, whereas the paired reduction provides the primary empirical evidence that padding suppresses the modeled gas channel.

Tasks with a median test-set size below 20 are reported as *underpowered* and excluded from the verdict. The criterion is applied independently to LR and RF; a task fails if either classifier satisfies both conditions, and the result tables report the classifier with the higher mean AUC.

The classifiers use the commit and access gas measurements, the only modeled policy-dependent channel remaining after transcript normalization.

6.2.2. Unsupervised clustering attacks

The clustering attacker drops the labels entirely and asks whether the gas features *naturally* group along the hidden property. We run K-Means (with K set to the number of classes, or K=2 for policy pairs) on the on-chain gas features, and score the recovered clusters against the true labels with ARI and NMI. Values near 0 mean the clusters carry no information about the hidden label; values near 1 mean full leakage.

PCH clusters all executions globally (no per-layer splitting), so the large-sample ARI/NMI are read directly. SPP is clustered the same way, once per annotation axis of Section 6.6 with K equal to that axis's bucket count, and its large-sample ARI/NMI are likewise read directly.

For PI, clustering is scored per pair, so each side of a pair must contribute enough executions; this requires two safeguards. **Balanced pairs.** The clustering uses a dedicated pair set built from the same executions: the policies with ≥ 30 executions each (the depth layer of Section 6.3) are paired into 127 balanced 30-vs-30 contrasts. The pair set of the supervised PI attack (Section 6.5) is unsuitable for per-pair clustering: most of its pairs hold a single execution per side, and its most imbalanced hold 1 execution against 30, so K-Means cannot recover a meaningful partition, and ARI/NMI ≈ 0 would reflect the degenerate setup rather than privacy. (The supervised attack itself is unaffected: it pools executions within a pair type rather than partitioning each pair.) **Permutation-debiased NMI.** Even after balancing, each side of a pair holds only 30 executions. At that scale the null simulation in Section 6.8 shows that finite-sample effects alone can raise NMI above zero while ARI stays at chance, so ARI is read directly and NMI must be debiased. The cluster assignments are computed once (K-Means does not see the labels) and the observed NMI is averaged over 20 K-Means seeds; the pair labels are then shuffled 200 times and the NMI recomputed against the fixed assignments on each shuffle, giving the null distribution of NMI under zero true signal. The debiased NMI is the observed NMI minus the mean of this null distribution, and the empirical p-value p_{NMI} is the fraction of shuffles whose NMI meets or exceeds the observed value. The cross-check flags a pair only if its debiased NMI is positive *and* significant ($p_{\mathrm{NMI}} < 0.05$).

6.3. Dataset and corpus generation

Shared corpus. We evaluate all three objectives on one policy corpus, so the supervised attacks and the clustering cross-check see the same transactions. The corpus holds 1077 distinct policies run as 1773 on-chain executions on *Polygon Amoy*, each execution being one commit transaction followed by one access transaction. The policies are pro-

duced by a deterministic generator with a fixed seed (`unified-decoupled-2026`). For reproducibility, this generator uses a deterministic test issuer to create the private profile-credential witnesses used by policies that exercise the credential predicate; this testing key does not model issuer governance or production key management.

For each policy the generator picks its content and its structure. The content is a set of DUO data-use terms drawn from ten terms: five ontology-bound terms, each checked against a code set from a biomedical vocabulary (DS against ICD-10, RS (research-specific restrictions) against the Ontology for Biomedical Investigations, OBI [40], POA against the Human Ancestry Ontology (HANCESTRO), GS (geographical restriction) against the ISO 3166 country codes, CC against SNOMED Clinical Terms (SNOMED CT)), and five profile-backed terms (GRU, HMB, NMDS, GSO, NCU) that are derived from the signed claims of the requester's profile credential (Section 5). The structure is the logical form joining the terms; whitelist sizes are drawn once per policy from a global set of buckets, independently of the logical form, so that the four structural axes of Section 6.6 are not functions of one another. For example, one generated policy permits research-specific and disease-specific use (DUO terms RS and DS) and grants access only when *both* conditions hold, joined by `AND`: the study design belongs to the OBI *observational and comparative designs* family, and the requested record falls in the ICD-10 *diseases of the circulatory system* chapter. Each condition is one predicate—hence the two predicates—and each is checked against a whitelist of the codes it admits: 8 OBI study-design codes and 8 ICD-10 codes drawn from that chapter, the "medium" parameter-size bucket of Table 7.

Every code set is a subset of its vocabulary, and the recorded whitelist size is the number of leaves actually committed. Table 7 gives the resulting counts for every label and structural bucket.

Two-layer execution design. We run the policies in two layers that serve different needs. The *coverage layer* runs 1053 policies once each. It provides breadth, spreading distinct policies across the content labels and structural buckets, so the PCH and SPP attacks have enough different policies to train on one set and test on another.

The *depth layer* runs 24 policies 30 times each (720 executions). It provides depth, which the PI attacks need because they compare two policies execution-by-execution, and for that each policy must have multiple executions of its own. This layer is what lets PI form its 127 balanced 30-vs-30 pairs and run the per-pair clustering check (Section 6.2).

Label restriction and statistical power. We restrict the generated content labels to those with enough support for a statistically powered attack: ten DUO data-use terms and the ten ICD-10 chapters, sampled so that the smallest class reaches at least 30 executions. The complete DUO has 23 data-use terms and ICD-10 has tens of thousands of leaf codes; generating and executing enough policies for every leaf term to reach adequate power would make the on-chain experiment infeasible. Category-level labels are therefore the finest granularity at which the attack has adequate power. They do not directly test individual leaf terms: a signal specific to one term could be diluted within its category, so a PASS at the category level does not rule out a term-specific leak. The transcript mechanisms themselves do not depend on the policy content: the verifier work and the public interface are fixed, and the padding band is the same for every policy (Section 5).

Collection. Executions were collected as live transactions on *Polygon Amoy* under the defense configuration of Section 6.2 (fixed circuit, commitment-only interface, randomized gas padding); observables are extracted from the resulting receipts, calldata, and event logs. All executions completed with the outcome the generator planned, and the feature file holds all of them. As established in Section 6.1, transcript normalization leaves gas as the only policy-dependent channel, so the residual per-transaction variation the attacker sees is confined to the two gas measurements.

Table 7. Composition of the shared policy corpus (generator seed `unified-decoupled-2026`). DUO terms are multi-label, so their counts overlap; ICD-10 rows cover only the ICD-bearing policies; each SPP axis partitions the full corpus.

Grouping	Category	Policies	Executions
DUO data-use term (multi-label; counts = policies/executions containing the term)			
	GRU	302	447
	DS	422	770
	RS	412	760
	POA	392	624
	GS	382	556
	CC	392	769
	NCU	206	380
	HMB	331	447
	NMDS	237	469
	GSO	236	352
ICD-10 chapter (ICD-bearing policies only)			
	ch0	44	131
	ch1	49	49
	ch2	33	62
	ch3	55	55
	ch4	41	70
	ch5	34	63
	ch6	48	135
	ch7	48	77
	ch8	33	62
	ch9	37	66
	ICD-bearing subtotal	422	770
SPP structureType			
	trivial	130	130
	and_only	294	526
	or_only	303	535
	and_or_mixed	350	582
SPP nestingDepth			
	flat	130	130
	shallow	210	413
	medium	335	509
	deep	402	721
SPP predicateCombo			
	single	257	344
	few	645	1080
	many	175	349
SPP paramType			
	none	147	234
	small	232	406
	medium	352	584
	large	346	549
Corpus total		**1077**	**1773**

6.4. Policy Content Hiding (PCH)

Property. PCH requires that the public-chain observer cannot recover the *content* of a policy—its DUO data-use terms or the ICD-10 disease codes DS references—better than chance.

Scope of the labels. The attack targets the ten DUO data-use terms of the corpus—general research use (GRU), disease-specific research (DS), research-specific restrictions (RS), population origins or ancestry research (POA), geographical restriction (GS), clinical care use (CC), non-commercial use (NCU), health or medical or biomedical research (HMB), no general methods research (NMDS), and genetic studies only (GSO)—and disease codes aggregated to the ten ICD-10 chapters. Section 6.3 explains this label restriction, its statistical-power rationale, and what category-level labels do not test.

Supervised attack. Each DUO term defines a one-vs-rest binary task, on which we run the multi-seed pipeline of Section 6.2. Table 8 reports the per-term verdicts: ev-

ery DUO task stays below the 0.55 AUC line, with mean AUC between 0.477 and 0.540; significant-seed fractions run up to 23% (GRU), short of the 50% gate, and with the AUC unmet none approaches the failure criterion.

Table 8. Policy Content Hiding: per-term supervised attack over 30 policy-disjoint seeds (one-vs-rest tasks; higher-scoring classifier per task; med n: median test-set size). All ten DUO tasks pass.

Term	AUC (mean $\pm$ std)	AUC gap (mean)	%sig	med n
DUO GRU	0.540 ± 0.026	0.041	23%	323
DUO DS	0.514 ± 0.029	0.027	3%	323
DUO RS	0.498 ± 0.028	0.022	0%	323
DUO POA	0.477 ± 0.024	0.027	0%	323
DUO GS	0.502 ± 0.027	0.023	0%	323
DUO CC	0.515 ± 0.029	0.025	3%	323
DUO NCU	0.528 ± 0.027	0.031	7%	323
DUO HMB	0.509 ± 0.027	0.024	3%	323
DUO NMDS	0.516 ± 0.030	0.025	3%	323
DUO GSO	0.502 ± 0.031	0.026	3%	323

The same pipeline applied to the ten ICD-10 chapters (one-vs-rest over the ICD-bearing rows only, median test n=127 per chapter; Table 9) passes nine chapters and fails one. Chapters ch1, ch2, and ch4 sit just above the AUC line (mean AUC 0.558–0.570) but reproduce in at most 23% of seeds, so they are boundary cases. Chapter ch5 (diseases of the nervous system) meets both legs: mean AUC 0.667 with 53% of seeds significant. We report it as a FAIL of the criterion and examine below what the classifier separates.

Table 9. Policy Content Hiding: per-chapter ICD-10 supervised attack over 30 policy-disjoint seeds (higher-scoring classifier per chapter; med n: median test-set size).

Chapter	AUC (mean $\pm$ std)	AUC gap (mean)	%sig	med n	Verdict
ICD ch0	0.472 ± 0.078	0.064	0%	127	PASS
ICD ch1	0.570 ± 0.066	0.075	23%	127	PASS
ICD ch2	0.565 ± 0.071	0.081	10%	127	PASS
ICD ch3	0.540 ± 0.061	0.061	7%	127	PASS
ICD ch4	0.558 ± 0.061	0.070	13%	127	PASS
ICD ch5	0.667 ± 0.069	0.167	53%	127	FAIL
ICD ch6	0.464 ± 0.077	0.065	0%	127	PASS
ICD ch7	0.525 ± 0.064	0.058	3%	127	PASS
ICD ch8	0.478 ± 0.078	0.063	0%	127	PASS
ICD ch9	0.511 ± 0.074	0.062	0%	127	PASS

What the ch5 classifier separates. The two gas measurements are the only non-constant observables. For each padded access transaction, the recorded gas can be decomposed as

$$\texttt{accessGasUsed} = \texttt{padTarget} + \texttt{residualOverhead},$$

where the TPS draws `padTarget` uniformly from the padding band (Section 5). On Polygon Amoy, the residual overhead ranges only from 4,661 to 4,750 gas across all 1,773 executions, so almost all variation in the padded access gas comes from the randomized target. The observer sees only the total gas, whereas the capture log also records each padding target. We therefore rerun the analysis on `accessGasUsed` $-$ `padTarget` to determine whether the ch5 separation remains after removing the randomized component. **This subtraction is diagnostic of the padded transcript only: the resulting residual overhead is not the gas usage of an unpadded execution.**

Across all 20 PCH tasks, the single-feature rank AUC of the padded access gas, computed over all executions without a classifier, coincides with that of the padding target to three decimal places, while the residual AUC lies between 0.47 and 0.52. For ch5, the gas and target AUCs are both 0.601, whereas the residual AUC is 0.472. The 34 ch5 policies happened to draw targets whose mean is 33,000 gas higher than that of the other 388 ICD-bearing policies, a 3.2σ deviation for a uniform draw of that size. Because each coverage-layer policy is executed once, the 30 evaluation seeds repeatedly partition the same fixed draws, causing this accidental difference to reproduce across seeds.

Applying the evaluation protocol to the residual gives a mean AUC of 0.556 with 10% significant seeds for logistic regression and 0.561 with 17% significant seeds for random forest, both yielding PASS. Applying it to the padding target alone reproduces the observed FAIL (0.686, 63%). Moreover, in the separate unpadded ablation, ch5 is already near chance (AUC 0.510 for logistic regression and 0.466 for random forest). Thus, for ch5 specifically, we find no detectable policy-dependent gas signal without padding; the FAIL in the padded corpus is instead caused by an accidental finite-corpus correlation between the random padding targets and the ch5 labels. This task-specific result does not imply that padding is unnecessary: the unpadded ablation reveals substantial leakage for many other policy contrasts, as shown in Section 6.9.

6.5. Policy Indistinguishability (PI)

Property. PI strengthens PCH: even when two policies are *semantically related*, the observer must not tell which one was executed.

We construct policy pairs of three types, defined by semantic similarity: `semantic_equivalent` (near-identical semantics, similarity $\geq$ 0.999, but a different parameter fingerprint—whitelist size, Merkle depth, or proof type), `weak_equivalent` ($\geq$ 0.95 semantic overlap), and `controlled_difference` (similarity $<$ 0.95, the least similar pairs by construction; reported for reference but excluded from the verdict).

Semantic similarity here is a fixed heuristic score over policy semantics—the Jaccard overlap [41] of DUO terms combined with agreement of the ICD-10, OBI, and other scope fields—that returns 1 when two policies share an identical normalized logic string. The $\geq$ 0.999 threshold therefore selects logically identical policies and $\geq$ 0.95 their close approximations. To build the pairs we score every candidate policy couple by this similarity, assign it to one of the three types above, and sample a fixed budget of 100 pairs per type, or every available pair when fewer exist (71 `weak_equivalent` pairs); each retained pair is an ordered couple (P_0, P_1) that the supervised attack below tries to tell apart.

Supervised attack. Each pair is an ordered couple (P_0, P_1), and the executed policy defines a binary label $b \in \{0, 1\}$; the attacker's task is to predict b from the on-chain features of a single execution, i.e. to decide which policy of the pair produced it. Indistinguishability is thus a two-class problem whose ideal outcome is a guess no better than a random guess, corresponding to a classifier AUC of 0.5.

We train a logistic-regression and a random-forest classifier and, per pair type, report the higher-AUC one, following the convention of Section 6.2. We evaluate this single-execution attacker over 30 policy-disjoint seeds (Table 10). Pooled over all pairs, the AUC is 0.509 ± 0.035 with 3% of seeds significant. Both gating pair types stay below the 0.55 line: `semantic_equivalent` at 0.549 (7% significant seeds) and `weak_equivalent` at 0.544 (10%), so PI passes. `controlled_difference`, the least similar pairs by construction and hence the easiest to separate, is excluded from the verdict and reported for reference; under the full mechanism it sits at 0.544 with 13% of seeds significant, whereas the ablation of Section 6.9 separates the same pair type at AUC 0.71 once padding is removed, which confirms that the pair construction produces a distinguishable positive control.

Table 10. Policy Indistinguishability: single-execution distinguisher over 30 policy-disjoint seeds, by policy-pair type (higher-scoring classifier per type; med n: median test-set size). `controlled_difference` is reported for reference and excluded from the PI verdict. All gating rows pass.

Pair type	Pairs	AUC (mean ± std)	AUC gap (mean)	%sig	med n
overall (all types)	271	0.509 ± 0.035	0.027	3%	162
semantic_equivalent	100	0.549 ± 0.064	0.070	7%	60
weak_equivalent	71	0.544 ± 0.079	0.079	10%	42
controlled_difference †	100	0.544 ± 0.064	0.062	13%	60

† Reported for reference; excluded from the PI verdict.

6.6. Structure & Parameter Privacy (SPP)

Property. SPP requires that neither the *shape* of a policy (logical structure, nesting depth, predicate coverage) nor the *size* of its parameters (whitelist/set sizes) leaks to the observer. We organize the test along four annotation axes—logical structure type (`structureType`), predicate count (`predicateCombo`), parameter size (`paramType`), and nesting depth (`nestingDepth`)—and test each pair of buckets within an axis.

`structureType` has four buckets: `trivial` (a single predicate with no boolean connective, e.g. policy logic `"DS"`), `and_only`, `or_only`, and `and_or_mixed`. `predicateCombo` buckets policies by predicate-namespace count into `single` (1), `few` (2–3), and `many` (≥ 4); `paramType` buckets by whitelist size into `none` (no whitelist: the policy is decided by profile-backed terms only), `small` (1–4), `medium` (5–16), and `large` (≥ 17); `nestingDepth` buckets by the maximum nesting of the normalized expression into `flat` (≤ 1), `shallow` (2), `medium` (3), and `deep` (≥ 4). The generator draws the whitelist size independently of the logical form, so the four axes are not functions of one another (Section 6.3); the per-bucket policy and execution counts are listed in Table 7.

Supervised attack. Table 11 reports all 21 pairwise contrasts over 30 policy-disjoint seeds, with the per-contrast AUC taken from the higher-scoring of the two classifiers following the convention of Section 6.2. Most contrasts sit at chance, with mean AUC between 0.47 and 0.54. Three exceed the 0.55 line without reproducing: `paramType` `none/small` (0.571, 40% significant seeds), `none/large` (0.568, 33%), and `nestingDepth` `flat/medium` (0.555, 27%); the two `none` contrasts separate the profile-only policies, whose 147 policies form the smallest minority class of the axis. One contrast, `nestingDepth` `medium/deep`, meets both legs of the criterion (mean AUC 0.563, 53% significant seeds) and is reported as FAIL.

Its separability has the same origin as the PCH ch5 case (Section 6.4). The single-feature rank AUC of the access gas for this contrast equals that of the randomized pad target alone (0.440 for both, i.e. deep policies drew lower targets) and the residual after subtracting the target scores 0.467; over all 21 contrasts the gas and target AUCs agree to three decimals, and the residual AUC stays within 0.47–0.56. Rerunning the protocol with the residual in place of the access gas returns the contrast to PASS (logistic regression 0.530 with 13% significant seeds; random forest 0.544 with 33%), while the target alone reproduces the FAIL (0.568, 57%). What remains above chance on this contrast is the per-policy random draw that the padding itself introduces, not a stable structural channel. Section 6.9 provides the corresponding positive control: when gas padding is removed, the same SPP attacker separates structural and parameter-size buckets with AUCs up to 0.90, and that signal collapses under the full normalization.

6.7. Unsupervised clustering: a label-free cross-check

The supervised attacks (Sections 6.4–6.6) assume an adversary who can train on labeled executions. A weaker, label-free attacker asks whether the on-chain gas features *cluster* along the hidden property with no supervision at all.

Table 11. Structure & Parameter Privacy: 21 pairwise contrasts across four annotation axes over 30 policy-disjoint seeds (higher-scoring classifier per contrast; med n: median test-set size).

Contrast	AUC (mean ± std)	AUC gap (mean)	%sig	med n	Verdict
structureType					
trivial / or_only	0.520 ± 0.048	0.043	0%	130	PASS
trivial / and_only	0.502 ± 0.040	0.033	3%	127	PASS
trivial / and_or_mixed	0.502 ± 0.042	0.034	0%	144	PASS
or_only / and_only	0.506 ± 0.034	0.027	0%	179	PASS
or_only / and_or_mixed	0.518 ± 0.035	0.032	10%	196	PASS
and_only / and_or_mixed	0.503 ± 0.040	0.032	7%	193	PASS
predicateCombo					
single / few	0.523 ± 0.035	0.033	13%	271	PASS
single / many	0.519 ± 0.048	0.041	7%	130	PASS
few / many	0.527 ± 0.048	0.045	17%	246	PASS
paramType					
none / small	0.571 ± 0.046	0.071	40%	114	PASS
none / medium	0.525 ± 0.035	0.035	3%	150	PASS
none / large	0.568 ± 0.038	0.069	33%	148	PASS
small / medium	0.530 ± 0.037	0.037	20%	175	PASS
small / large	0.504 ± 0.036	0.028	3%	173	PASS
medium / large	0.493 ± 0.027	0.022	0%	209	PASS
nestingDepth					
flat / shallow	0.522 ± 0.038	0.036	3%	102	PASS
flat / medium	0.555 ± 0.032	0.055	27%	140	PASS
flat / deep	0.468 ± 0.046	0.047	3%	160	PASS
shallow / medium	0.521 ± 0.041	0.039	7%	164	PASS
shallow / deep	0.533 ± 0.036	0.039	17%	184	PASS
medium / deep	0.563 ± 0.028	0.063	53%	221	FAIL

Table 12. Unsupervised clustering attack (K-Means, K = number of classes, K=2 for pairs) on the on-chain gas features, with the permutation p-value of ARI. PCH clusters all executions globally.

Property	Target	n	#cls.	ARI	p_{ARI}	NMI
PCH	DUO terms	1773	10	0.0000	0.48	0.0109
PCH	ICD-10 chapters	770	10	0.0021	0.06	0.0281
PI	policy pairs (30v30)	127	2	0.0020	—	0.0139
SPP	`structureType`	1125	4	0.0004	0.33	0.0034
SPP	`predicateCombo`	1125	3	−0.0005	0.58	0.0015
SPP	`paramType`	1125	4	−0.0000	0.49	0.0030
SPP	`nestingDepth`	1125	4	0.0071	0.005	0.0088

Table 12 consolidates this attack across all three objectives: K-Means recovers essentially nothing, with ARI within ±0.003 and NMI below 0.03 for every DUO, PI, and SPP target except one. NMI values are interpreted against the null simulation in Section 6.8: the slightly higher ICD-10 NMI (0.0281) lies within the range of a signal-free ten-class clustering with the same chapter sizes ($p = 0.15$), and permuting the chapter labels against the observed clusters gives the same reading (debiased NMI 0.0044, $p = 0.08$). The exception is the SPP `nestingDepth` axis, whose ARI of 0.007 is tiny in magnitude but permutation-significant ($p_{\text{ARI}} = 0.005$). It is the unsupervised face of the `medium/deep` contrast of Section 6.6: clustering the same executions on the residual gas (access gas minus the logged pad target) returns the axis to ARI −0.0004 ($p = 0.60$), while every other axis is unchanged, so the grouping follows the per-policy random draws of the padding. With that qualification, the unsupervised attacker agrees with the supervised verdict as a second, label-free attack family on the same dataset (Figure 2).

Removing sample-size confounds. The PI clustering inputs are rebalanced into 30-vs-30 contrasts and permutation-debiased as described in Section 6.2, so the near-zero ARI/NMI is not a degenerate-measurement artifact: over the 127 balanced pairs the mean ARI is 0.002, the mean debiased NMI 0.001, and 6 pairs (5%) reach nominal significance, the fraction expected under the null at $\alpha = 0.05$. The conclusion matches the supervised PI

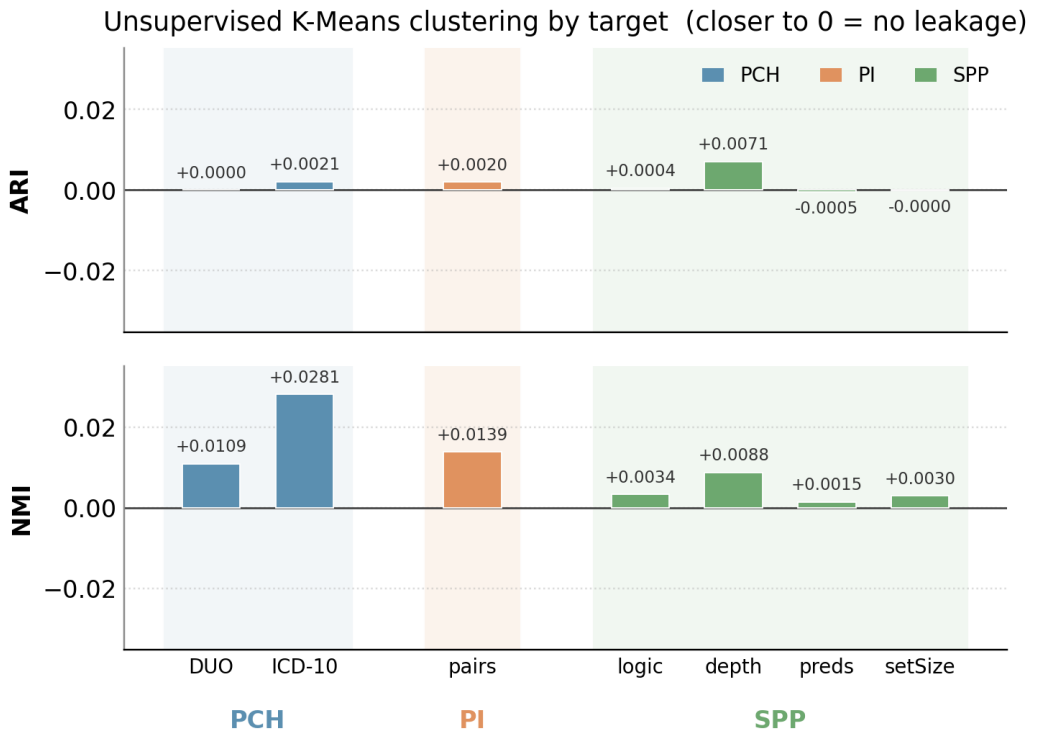


Figure 2. Unsupervised K-Means clustering across all three objectives: ARI (top) and NMI (bottom) per target.

verdict (Section 6.5), and the ablation (Section 6.9) confirms the clustering device itself is live: on the un-padded transcript the same balanced pairs cluster with a mean debiased NMI 0.67 higher, and the signal disappears after normalization.

6.8. Sample-size sensitivity of NMI

The clustering cross-check (Section 6.7) treats ARI as the primary conservative indicator because NMI can take moderately high values under small samples. To justify that choice, we quantify how *finite-sample effects* alone can inflate NMI even when there is *no* discriminative signal, using an idealized null in which all classes are statistically indistinguishable.

Null hypothesis. We sample every class from the *same* distribution, $z_i \sim \mathcal{N}(0, I_d)$ for all classes, where I_d is the d-dimensional identity covariance and $d = 2$ matches the two gas features the clustering uses. Since no class carries class-specific information, a clustering algorithm should not recover class membership, and both NMI and ARI should concentrate around 0 as the sample size grows.

Simulation. For each configuration we run $T = 1000$ independent trials. In each trial we (i) generate G classes with n i.i.d. samples per class from $\mathcal{N}(0, I_d)$, for $N = Gn$ samples in total, assigning each sample a class label by construction order (these labels are independent of the sample values, since all classes share one distribution), (ii) standardize the pooled data and apply K-Means with $K = G$ and ten initializations, as in the evaluation, and (iii) score the predicted assignments against these assigned class labels with NMI and ARI. We sweep n while holding G fixed, and report the trial means $\overline{\text{NMI}}$ and $\overline{\text{ARI}}$. We consider a **PI-like** binary structure ($G = 2$, $n \in \{3, 10, 50, 100\}$), a five-class structure ($G = 5$, $n \in \{3, 10, 100\}$), and a ten-class structure ($G = 10$, $n \in \{3, 10, 77\}$) together with the observed ICD-10 chapter sizes (49–135 executions per chapter, $N = 770$).

Results. Table 13 shows that, although all classes are indistinguishable by construction, **NMI is noticeably positive at very small n but collapses toward zero as n grows**, whereas

ARI stays near zero throughout. For the binary structure, $\overline{\text{NMI}}$ falls from 0.2074 at n=3 (N=6) to 0.0420 at n=10, and to 0.0072/0.0034 at n=50/100. The same trend holds with more classes, at a higher level for every n: with five classes $\overline{\text{NMI}}$ goes from 0.4152 (n=3) to 0.0102 (n=100), and with ten classes from 0.5346 (n=3) to 0.0239 (n=77). For the observed ICD-10 chapter sizes, averaging each trial over 20 K-Means seeds as the evaluation does, the null NMI has mean 0.0246 and 95th percentile 0.0302; the observed ICD-10 NMI of 0.0281 is reached in 145 of 1000 null trials ($p = 0.15$). Hence moderately high NMI in small-sample clustering should not be read as actionable leakage, and ARI is the more conservative indicator of practical recoverability in our unsupervised evaluation.

Table 13. Sample-size sensitivity of NMI and ARI under the null hypothesis where all classes are i.i.d. sampled from $\mathcal{N}(0, I_d)$ (T=1000 trials per row; G classes, $N = Gn$ samples).

Structure	n (per class)	G	N	$\overline{\text{NMI}}$	$\overline{\text{ARI}}$
PI-like	3	2	6	0.2074	0.0127
PI-like	10	2	20	0.0420	0.0008
PI-like	50	2	100	0.0072	−0.0003
PI-like	100	2	200	0.0034	−0.0004
5-class	3	5	15	0.4152	−0.0034
5-class	10	5	50	0.1226	0.0003
5-class	100	5	500	0.0102	−0.0000
10-class	3	10	30	0.5346	−0.0009
10-class	10	10	100	0.2087	0.0003
10-class	77	10	770	0.0239	−0.0001
ICD-10 sizes	49–135	10	770	0.0244	−0.0001

6.9. Ablation: gas-normalization positive control

The results show that the mechanism hides content, policy identity, and structure/-parameters up to the two pad-target coincidences discussed above. The ablation asks *why*: how much of the residual attack surface is removed by randomized gas padding after the fixed circuit and commitment-only interface have already eliminated calldata-length, proof-size, and event-log channels.

We compare two aligned protection profiles on the same selected policy set. **B1** uses the fixed circuit, commitment-only interface, and TPS execution profile without commit/access gas padding. **B2** is the full mechanism used in the main evaluation, with randomized gas padding. The subset is chosen deterministically from the of-record corpus so that every content label and structural bucket keeps at least 70 executions and 8 independent policies and every depth-layer policy is retained: 1119 executions of 423 policies.

B1 replays exactly these policies (same tokens, randomizers, and commitments) without padding, so B1 and B2 differ only in the padding calls; B1 was executed three days after B2, on a fresh contract deployment, which the paired analysis treats as a non-contemporaneous control. This is the positive control for the gas channel: calldata length, proof size, and event-log shape are already fixed.

The paired leakage metric is

$$\Delta_{\text{leak}} = |\text{AUC}_{\text{B1}} - 0.5| - |\text{AUC}_{\text{B2}} - 0.5|, \tag{16}$$

where positive values mean padding moved the attacker closer to chance; we bound each Δ_{leak} with a 95% stability interval over 30 paired train/test splits (denoted CI in Table 14

and Figure 3). Table 14 is intentionally separated from the main privacy verdicts: the main verdicts answer whether the full mechanism is private; this ablation attributes that privacy to a specific defense layer and reports its gas cost.

Table 14. Gas-padding ablation on Polygon Amoy: B1 (no padding) versus B2 (deployed randomized padding) on the same 1119-execution subset; supervised rows are the largest-signal contrasts per experiment (random-forest AUC).

Attack	Target	B1 (unpad.)	B2 (pad.)
Supervised (RF AUC; Δ_{leak} with 95% interval)			
SPP	`structureType trivial/and_or_mixed`	0.902	0.490
	$\Delta_{\text{leak}} = 0.347$ CI$[0.334, 0.360]$		
SPP	`predicateCombo single/few`	0.844	0.494
	$\Delta_{\text{leak}} = 0.301$ CI$[0.287, 0.314]$		
PCH	DUO HMB	0.876	0.495
	$\Delta_{\text{leak}} = 0.337$ CI$[0.319, 0.352]$		
PI	`controlled_difference` †	0.676	0.446
	$\Delta_{\text{leak}} = 0.088$ CI$[0.062, 0.111]$		
Unsupervised (NMI, B1−B2 with policy-group bootstrap 95% interval)			
PCH	DUO terms	0.349 [0.316, 0.407]	
SPP	logic structure	0.102 [0.018, 0.138]	
PI	balanced pairs (debiased NMI)	0.670 [0.591, 0.743]	
Gas cost (mean over the subset, B1 → B2)			
Commit gas		219,737	304,098 (+38.4%)
Access gas		723,462	1,056,674 (+46.1%)

† Positive control: the least similar pairs by construction, excluded from the PI verdict.

Padding removes otherwise-recoverable signal. Under the criterion of Section 6.2, the unpadded transcript fails 9 of the 10 DUO tasks, 14 of the 21 SPP contrasts, and the PI positive control, with mean AUCs up to 0.90; the padded transcript of the same subset passes every task.

Figure 3 reports the full set of 45 supervised contrasts: each point places a contrast at its unpadded deviation from chance $|\text{AUC}_{\text{B1}} - 0.5|$ against its padded deviation $|\text{AUC}_{\text{B2}} - 0.5|$, so points below the diagonal are contrasts where padding moved the attacker back toward chance. Every one of the 21 contrasts on which B1 carries a sizable signal ($|\text{AUC}_{\text{B1}} - 0.5| > 0.15$) collapses below that line under B2; across the 45 random-forest contrasts the median unpadded deviation of 0.137 drops to 0.038, and the per-family median Δ_{leak} is $+0.173$ (SPP), $+0.058$ (PCH), and -0.011 (PI, whose gating pair types are already at chance under B1).

Source of the unpadded signal. The B1 result deserves a precise reading, because the fixed circuit already makes the policy logic invisible to gas: the Groth16 verifier performs the same work for every proof, and the access-control path is the same for every policy. The unpadded access gas of the 1,119 executions spans only 723,288–723,564 gas (one cold-start first call at 741,276 aside), in steps of 12 gas, and the ten ICD-10 chapter tasks do not separate on it.

Twelve gas is the price difference between a zero and a non-zero calldata byte set by Ethereum Improvement Proposal 2028 (EIP-2028), and decoding the calldata of the transactions shows where the variation comes from: the requester pseudonym id_{req}. In the evaluation, a requester who presents a profile credential is identified by a Poseidon-derived pseudonym that fills the 256-bit word, whereas a requester of an ontology-only policy carries a 128-bit random pseudonym whose upper 16 bytes are zero; the 192-gas difference between the two encodings is exactly the gap between the two clusters of B1

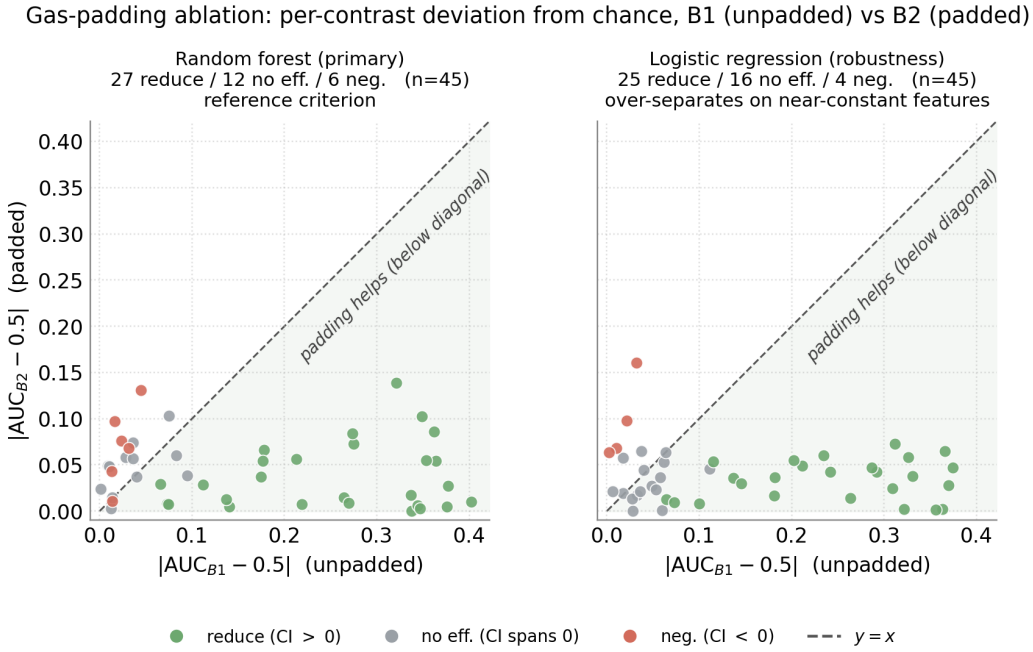


Figure 3. Gas-padding ablation over all 45 supervised contrasts: unpadded versus padded deviation from chance, colored by the sign of the 95% interval of Δ_{leak}.

gas values, and whether a policy activates the profile predicate is correlated with its DUO terms, its structure type, and its parameter bucket. The remaining ± 84 gas come from the zero-byte counts of the proof elements, the policy and request identifiers, and the epoch, none of which depends on the policy.

The unpadded leak is therefore not the policy logic but a sub-kilogas, deterministic calldata-pricing artifact of how one public transaction argument is encoded; it is a real channel for the observer nonetheless, and it is the class of channel that padding is designed to absorb, because randomizing the target by $\pm 10\%$ ($\pm 105{,}000$ gas) submerges any deterministic difference of a few hundred gas regardless of its origin. Encoding every pseudonym at full width would close this particular channel without padding; randomized padding remains the more general defense because it also absorbs comparable deterministic gas differences caused by other public encodings or execution effects.

As summarized in Table 14, the clustering attacker shows the same pattern on a second, label-free attack family: the debiased NMI of the balanced PI pairs is 0.67 higher under B1 than under B2, and the DUO-term and logic-structure clusterings lose 0.35 and 0.10 of NMI, each with a policy-group bootstrap interval excluding zero.

The effect is conditional, and we report it as such: the 12 random-forest contrasts with no detectable effect sit in the lower-left of the figure, where B1 is already near chance, and the 6 contrasts on which padding *raises* the deviation ($\Delta_{\text{leak}} < 0$, red) are cases where B1 is at chance and the padded transcript's remaining deviation is the per-policy pad-target draw examined in Sections 6.4 and 6.6 (ICD ch5 and `nestingDepth medium/deep` among them). These per-contrast counts are descriptive, not multiplicity-corrected; the causal claim rests on the direction and magnitude of the shift where a signal is present, corroborated by the independent clustering attacker.

Cost. Padding is not free: over the subset it raises mean commit gas from 219,737 (B1) to 304,098 (B2), a 38.4% increase, and mean access gas from 723,462 to 1,056,674, a 46.1% increase. The padded access cost is set by the padding band: the TPS draws the target uniformly from 945,000–1,155,000 gas, and the base of that band is chosen above the bare access cost measured on the target network (748,000 gas on Amoy) so that the whole band

stays above real usage. At the 30 gwei tip used throughout, one padded access transaction costs about 0.032 POL (the native token of Polygon) and one commit about 0.009 POL; the overhead is therefore a fixed per-transaction premium of roughly 330,000 gas on access and 85,000 on commit, independent of transaction volume, and it does not grow with policy complexity, because neither the bare cost nor the band depends on the policy.

Table 15 gives the single-point cost characterization of the evaluated prototype configuration at DEPTH $= 64$. The circuit compiles to a rank-1 constraint system (R1CS) of 81,740 constraints; across the 1,773 Polygon Amoy executions, mean witness and proving times on the two-core capture host are 763.6 ms and 2,884.2 ms, respectively. Gas, logs, calldata lengths, and transaction objects are taken from the same executed dataset, whose transactions use the context-bound five-signal statement.

Table 16 reports two offline capacity sweeps, each with 30 verified proofs per configuration on an Apple M1 using rapidsnark. With $N = 13$, $M = 11$, and eight Merkle levels fixed, increasing DEPTH from 32 to 128 raises the circuit from 64,945 to 134,009 constraints and the median proving time from 0.411 to 0.734 s. With DEPTH $= 64$ fixed, increasing the ontology-tree depth from 4 to 16 raises the circuit from 73,876 to 97,468 constraints and the median proving time from 0.512 to 0.571 s for a five-slot ontology-heavy policy. Every variant retains five public signals, so enlarging either capacity changes the prover artifacts but not the verifier interface.

Table 17 evaluates policy content within the evaluated circuit (DEPTH $= 64$, eight Merkle levels). Across simple, ontology-heavy, profile-heavy, and max-mixed policies, median witness time is 0.335–0.354 s and median proving time is 0.484–0.545 s, despite the encoded programs ranging from 3 to 61 tokens and activating from one to all 11 production predicate slots. The larger end-to-end time for profile-backed cases reflects off-circuit credential and allowlist preparation; the fixed-circuit witness/prove/verify work remains stable. Under load, throughput stays between 0.79 and 0.94 proofs/s as concurrency rises from one to four, while median latency grows from 1.2 to 4.1 s; Section 7 converts the measured capacity into requests per day.

Table 15. Single-point cost profile of the evaluated prototype configuration at DEPTH $= 64$. Gas, logs, calldata, and local prover timings are summarized over the 1,773 revised Polygon Amoy executions.

Metric	Value / source
Circuit size	81,740 R1CS constraints; 80,884 wires; DEPTH $= 64$
Public interface	4 public inputs plus 1 output
Padded on-chain gas	Commit 304,167; access 1,057,283 mean gas
Event logs	Commit: 1 log / 180 bytes; access: 2 logs / 296 bytes
Calldata length	Commit 196 bytes; access 612 bytes (constant across policies)
Local witness time	763.6 ms mean
Local prove time	2,884.2 ms mean

7. Discussion

Privacy findings. The evaluation supports a scoped claim: against the modeled public-chain metadata adversary, TPS-ZKPI provides empirical resistance to policy leakage. Of the 45 supervised tasks, 43 stay below the multi-seed leakage criterion, and unsupervised clustering shows no recoverable structure beyond one small, permutation-significant nesting-depth ARI.

The two tasks that meet the criterion—one ICD-10 chapter and the `nestingDepth medium/deep` contrast—are reported as failures, and both are carried by the random pad target: subtracting the logged target from the access gas returns each to PASS, and replacing the gas features by policy-independent draws reproduces such failures in four of ten

Table 16. Offline circuit-capacity sweeps (30 verified proofs per row). Timing cells report median (p95) in ms; every variant keeps $N = 13$, $M = 11$, and five public signals. WASM and zkey are the sizes of the witness generator and the proving key.

Sweep	Capacity	Constraints	Wires	WASM (MB)	zkey (MB)	Witness	Prove	Verify
Program capacity: Merkle levels fixed at 8; mixed policy								
DEPTH	32	64,945	64,441	8.6	38.7	285 (343)	411 (562)	11 (16)
DEPTH	64	81,740	80,884	8.7	52.2	350 (414)	521 (761)	11 (22)
DEPTH	128	134,009	132,449	10.1	87.5	535 (682)	734 (984)	11 (15)
Ontology capacity: DEPTH $= 64$*; five-slot ontology-heavy policy*								
Merkle levels	4	73,876	73,052	8.6	48.3	352 (415)	512 (605)	11 (17)
Merkle levels	8	81,740	80,884	8.7	52.2	347 (354)	499 (539)	11 (16)
Merkle levels	12	89,604	88,716	8.7	56.0	358 (395)	552 (632)	10 (15)
Merkle levels	16	97,468	96,548	8.7	59.9	370 (456)	571 (713)	10 (13)

Table 17. Policy-content and workload scalability of the evaluated circuit (DEPTH $= 64$, eight Merkle levels) on one Apple M1 host using rapidsnark.

(a) Policy content: 30 *verified proofs per policy*

			Time (ms): median (p95)			
Policy	Tokens	Slots	Witness	Prove	Verify	End-to-end
Simple	3	1	354 (418)	545 (746)	12 (14)	925 (1170)
Ontology-heavy	19	5	349 (391)	506 (550)	10 (12)	880 (939)
Profile-heavy	31	1	337 (349)	492 (522)	10 (11)	1479 (1516)
Max-mixed	61	11	335 (346)	484 (537)	10 (10)	1468 (1538)

(b) Load: 100 *verified requests per concurrency level, rotating the four policies of (a)*

Concurrency c	Throughput (proofs/s)	Latency (ms): median (p95)
1	0.805	1230 (1659)
2	0.787	2492 (3805)
4	0.936	4113 (5854)

Note: Tokens: program length ℓ (at most DEPTH $= 64$); slots: external-predicate slots the policy uses (of $M = 11$). End-to-end time also includes off-circuit evidence preparation. Latency is each request's processing time while up to c requests run at once; queueing is excluded. All timings follow five warm-up requests.

replicates (Sections 6.2–6.6). Such false failures are a property of the criterion on a corpus in which coverage-layer policies are executed once, so the thirty seeds resample the same random draws; we keep the criterion as stated and document the diagnostic, since lowering the significance threshold or raising the AUC threshold to absorb these cases would weaken it for genuine leaks.

The gas-padding ablation (Section 6.9) attributes the gas-channel protection to randomized gas padding: without it the same attacks recover the policy family with AUCs up to 0.90. Content hiding itself rests on the commitment-only interface and the content-independent transcript normalization; padding is the defense-in-depth layer that absorbs whatever deterministic gas differences remain, of which the ablation exhibits one that does not come from the policy logic at all but from the byte encoding of a public transaction argument. Together with fixed-length policy representation, reusable interpretation, and commitment binding, these mechanisms limit the extent to which public-chain observers infer hidden policy content or structural properties from authorization transactions.

The cost of padding. Within the evaluated scope, the residual cost is the randomized padding's gas overhead: on the ablation subset, about 46% on access gas and 38% on commit gas (Section 6.9). The overhead is a fixed premium per transaction—roughly 330,000 gas on access and 85,000 on commit at the band used here—so it scales linearly with transaction volume and does not compound: at the 30 gwei tip of the evaluation, a padded access transaction costs about 0.032 POL against 0.022 POL unpadded. Both the bare cost and the band are policy-independent: the fixed circuit gives every proof the same verifica-

tion work and the same five public signals, and the prover-side cost of larger capacities or different policy contents (Tables 16 and 17) is paid off-chain.

The offline load benchmark in Table 17 rotates the four policy classes through 100 verified requests after five warm-ups at each concurrency level. On the evaluated Apple M1, the best observed batch throughput is 0.936 proofs/s at concurrency four, with median and p95 request latencies of 4.113 and 5.854 s; CPU contention makes the concurrency-two result slightly lower, so these are single-host measurements rather than a linear-scaling claim. At this throughput, one prover handles about 80,900 requests per day, or about 53,900 if half of its capacity is held in reserve; these figures describe the measured host, not hospital traffic or public-chain throughput. On-chain, each padded access costs 1,056,674 gas on average (about 0.032 POL at 30 gwei), so the access cost grows linearly with the number of requests, and each policy commit adds a one-time 304,098 gas (0.009 POL).

The band must be positioned above the bare cost on the target network, which differs between networks (the same verifier costs 243,000 gas on a local EVM and 372,000 on Amoy, whose BN254 precompiles are priced higher); a deployment therefore calibrates the base target once per network, and the width of the band, $\pm 10\%$ here, is what determines how large a deterministic difference it can submerge. Padding cannot, of course, hide a difference larger than its band, and it does not address channels that are not gas.

TPS trust and credential provenance. The TPS trust assumption is the main architectural tradeoff. Moving confidentiality-sensitive proof generation to the TPS keeps policy-bound proving information away from requesters and public smart contracts while preserving on-chain verifiability, but it shifts part of the confidentiality burden to the service layer (Section 4). This tradeoff leaves on-chain verifiability intact but does affect liveness and accountability: the TPS can refuse to produce a proof, and such a refusal may be indistinguishable from a legitimate denial at the on-chain interface.

Attribute origin is only partially delegated to the TPS. For the profile-backed DUO terms, the circuit verifies the EdDSA-Poseidon signature described in Section 5 and derives eight policy attributes from the seven signed claims. A compromised TPS cannot modify the signed claims, policy commitment, requester pseudonym, resource, validity interval, or credential identifier without invalidating the signature. The implemented relation rejects a non-committed issuer, an invalid signature under the committed key, an expired or premature credential, and substitutions of any signed field. Other evidence has a narrower meaning: ontology proofs establish that a supplied code belongs to a provider-committed set, and requester allowlists establish that a holder pseudonym belongs to a provider-committed list; they do not authenticate every underlying real-world fact. Moreover, because the TPS receives the holder secret and credential witness in the current workflow, holder binding does not protect against a malicious TPS reusing otherwise valid evidence within its signed scope.

Issuer governance and deployment path. The remaining trust anchor is therefore issuer governance. The prototype binds the proof to a policy-selected issuer-key commitment, but it has no independent registry establishing that this key belongs to an accredited healthcare authority, and it does not check credential revocation. The evaluation uses a deterministic test-only issuer key to exercise the complete signature path; it is not evidence of institutional issuance or production key management.

A concrete deployment can add issuer governance without changing the Boolean interpreter: a governed registry would publish schema-qualified issuer keys and their validity epochs, while an issuer would sign a domain-separated credential containing a holder-key commitment, claim vector, authorization scope, validity interval, and unique credential identifier. The circuit would verify the credential signature and a Merkle proof that the issuer key is active in the registry, while continuing to derive policy attributes

from the signed claims. The holder would authorize each request by signing the context digest and request epoch, preventing a TPS that observes one credential presentation from generating proofs for fresh sessions; an epoch-specific sparse-Merkle or accumulator root would provide a private non-revocation witness for the credential identifier.

The registry and revocation roots can occupy fixed-format public or committed parameter slots, preserving the Boolean interpreter and the shape of the policy-independent public interface, although these checks require a revised circuit and new proving and verification keys. Existing healthcare credentials using Ed25519 or the Elliptic Curve Digital Signature Algorithm (ECDSA) require either higher-cost in-circuit verification or an additional SNARK-friendly credential; TPS re-signing cannot establish issuer-origin assurance.

Policy and context binding. As described in Section 5, the policy commitment covers both the padded encoding P_{enc} and its active length ℓ. Because ℓ also determines how many tokens the interpreter evaluates, an authorizing proof for h cannot be based on a shorter valid prefix. This integrity guarantee is separate from our privacy evaluation, which measures whether public-chain artifacts reveal information about the hidden policy.

The authorization context is likewise part of the proved relation. The circuit constrains the public digest d_χ to the requester identifier, resource, and nonce used in the witness, and the ACM recomputes the same digest from the transaction parameters. Consequently, changing any of these values invalidates the request.

The guarantee is scoped to the bound tuple within one ACM deployment. The digest does not include the transaction submitter or a chain/contract domain, so exact-context front-running and cross-deployment reuse require separate deployment-level controls. Moreover, id_{req} is a pseudonymous authorization identifier and is not, by itself, proof of a real-world identity or of the origin of every private attribute.

Future directions. Three directions follow. First, adding an explicit chain/contract domain and an authenticated submission channel would extend context binding beyond a single deployment and reduce front-running risk. Second, accountable proof obligations—for example, on-chain commitments to serve requests together with timeout-with-evidence mechanisms—would make proof refusals attributable. Third, the reduced-trust TPS instantiations of Section 4 (collaborative-SNARK or TEE-backed proving) would relax the trust concentration itself, including the attribute-origin limitation; realizing and evaluating them is left to future work.

8. Conclusion

This work presents TPS-ZKPI, a Trusted Proof Service-assisted zero-knowledge approach for policy-private consent enforcement in blockchain-based healthcare data sharing. TPS-ZKPI commits hidden provider-defined policies, evaluates bounded Boolean policies through a reusable interpreter, and normalizes the public verification transcript to reduce metadata-level leakage. Its authorization relation binds each proof to the committed policy and request context, while profile-backed attributes derive in-circuit from issuer-signed credentials under an explicit issuer-governance assumption.

The evaluation on Polygon Amoy provides empirical support for resistance to policy leakage under the modeled public-chain metadata adversary. Most supervised tasks remained below the multi-seed leakage criterion, and unsupervised clustering recovered no usable policy structure beyond one small nesting-depth result. Diagnostic analysis associated the two threshold crossings with random padding targets and reproduced them with policy-independent controls. The paired ablation establishes the role of randomized

gas padding: without it, the attacks recover policy-family signal; with the full mechanism, padding suppresses this channel at a substantial fixed per-transaction gas premium.

These findings are empirical and scoped to passive observation of commitments, proofs, public parameters, calldata, events, and gas on the generated-policy corpus. Offline scalability measurements cover bounded program and ontology capacities, fixed-circuit policy classes, and single-host proof throughput; they do not establish production hospital capacity. Policy confidentiality remains contingent on the TPS following its workflow, and institutional credential trust remains contingent on issuer governance. The proved policy class covers access-time checks for bounded Boolean policies over standardized terms. Outside this scope lie downstream purpose limitation, post-access duties, delegation, credential revocation, and channels involving timing, ordering, mempool activity, off-chain communication, adaptive probing, repeated access attempts, or correlations between authorization outcomes and policy classes, which the corpus does not exercise because every evaluated request is granted; the results do not establish policy confidentiality against these channels.

Three directions follow: reduced-trust proof generation and holder-signed request contexts can reduce the TPS trust concentration; governed issuer registries with in-circuit membership and revocation witnesses can strengthen credential provenance; and evaluation against active adversaries can test transcript normalization in the timing and ordering channels beyond the present model.

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Abbreviations

The following abbreviations are used in this manuscript:

TPS-ZKPI	Trusted Proof Service-assisted Zero-Knowledge Policy Interpreter
TPS	Trusted Proof Service
ACM	Access-Control Manager
ZKP	Zero-Knowledge Proof
SNARK	Succinct Non-interactive Argument of Knowledge
TEE	Trusted Execution Environment
RPN	Reverse Polish Notation
PCH	Policy Content Hiding
PI	Policy Indistinguishability
SPP	Structure and Parameter Privacy
DUO	Data Use Ontology
ICD-10	International Classification of Diseases, 10th Revision
OBI	Ontology for Biomedical Investigations
GA4GH	Global Alliance for Genomics and Health
CP-ABE	Ciphertext-Policy Attribute-Based Encryption
PRE	Proxy Re-Encryption
AUC	Area Under the (ROC) Curve
ARI	Adjusted Rand Index
NMI	Normalized Mutual Information

LR	Logistic Regression
RF	Random Forest
ABE	Attribute-Based Encryption
ABAC	Attribute-Based Access Control
MPC	Multi-Party Computation
HE	Homomorphic Encryption
SSI	Self-Sovereign Identity
IoMT	Internet of Medical Things
XACML	eXtensible Access Control Markup Language
SAML	Security Assertion Markup Language
EdDSA	Edwards-curve Digital Signature Algorithm
ECDSA	Elliptic Curve Digital Signature Algorithm
EVM	Ethereum Virtual Machine
EIP	Ethereum Improvement Proposal
R1CS	Rank-1 Constraint System
MONDO	Mondo Disease Ontology
HANCESTRO	Human Ancestry Ontology
SNOMED CT	SNOMED Clinical Terms

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